Appendix to "A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it"

Draft 2 (2026-10-21), for `docs/paper/MAIN_DRAFT2.md`. Every number is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body. "Registered" = a pre-registered decision or declared secondary; "(analysis)" = computed on stored streams or checkpoints, eval-only, never changing a verdict; its script is listed in App. O. App. E states and proves every theory statement the main text uses; statements not used are in `docs/paper/APPENDIX_theory_v3.md`. Delete this paragraph before submission.

Contents. A: models, recipes, batches. B: floors. C: per-seed tables, SOLVED-threshold sensitivity, evaluation scale. D: constructions. E: theory. F: exchange-rate details. G: rank details. H: sign details. I: CARoPE, CoPE and Selective RoPE. J: text world. K: what and where; the leak. L: PoPE's score. M: loops and depth on a rank-2 code. N: EM's recency deficit, a contrast case. O: analysis scripts.

A. Models, recipes, batches

A1. Architecture

component	setting
layers	1 (Sec. 6: 1-4 real layers and one block looped x4; ROBUST torus: 2 layers, d_model 256)
heads, width	2 heads, d_model 128, d_head 64; feed-forward 4 x d_model with GELU; LayerNorm before attention and before the feed-forward; dropout 0.1 on attention weights, attention output and feed-forward output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (content Q and K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias; r = 4 shared across heads unless stated (MapFormer's latent is per head); per-head variants: one W_in / W_out block per head; block-diagonal rank 4: the shared rank-4 latent with the off-diagonal blocks of W_out held at zero (each head reads rank 2)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = grid size (torus 64, ring 32, line 256); omega is a free parameter, its sign unconstrained
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	about 204k for the one-layer torus MapWM; 798,741 for four index layers; 203,925 for the looped block (`FACTORIAL_RESULTS.md`); 1.59M for the ROBUST torus model

Step parameterisations (`model.py`, `model_sign.py`, `model_wrap_exist.py`): signed (`Signed_r4` = `Vanilla_r4`), |.| (`Abs_r4`; the absolute value is applied after W_out on each of the 64 channel steps, adding no parameters), softplus (`Pos_r4`; softplus(0) = 0.693, so every token starts at about the same step), observation-step-free (`SignedAO_r4`, `AbsAO_r4` on the ring,

`SignedA04_r4`, `AbsA04_r4` on the torus: every non-move token's step is 0 by construction). Index RoPE (`RoPE`): canonical ladder base^(-2c / d_head), base 10000.

A2. Recipes

	torus (Secs. 5-8)	**ring and line (Secs. 6, 7)**	**Dyck-2 (Sec. 6)**
optimiser	AdamW, weight decay 0.05	AdamW, weight decay 0.05	AdamW, weight decay 0.01
peak learning rate	1e-3	1e-3 (ROBUST and LRSWEEP vary it)	1e-4
schedule	5% linear warm-up, cosine to 10% of the peak	same	same
batch	16 walks of T = 1024 moves (2,048 tokens); T = 128 batches of 128	128 walks of T = 128	128 sequences
budget	98 batches per epoch, 900 epochs (1800 in Sec. 7.2; 300 at T = 128)	98 batches per epoch, 300 epochs	560,000 sequences (1x) or 1,680,000 (3x)
loss	cross-entropy on revisited observations	same	next-token cross-entropy on every position
training map	one per run	redrawn for every walk	--
evaluation	100 walks on one held-out map, eval mode	400 walks, maps redrawn	L32 D12 test set

Run classes (`stats_core.classify_run`, per-epoch training loss): SOLVED if the mean of the final 5% of epochs is below 0.05; otherwise STALLED if the final-10% mean is within 5% of the preceding 10% mean, else DESCENDING. Continuations of constructed models (App. D): ring 300 epochs at lr 1e-3 (schedule re-warmed, fresh optimiser state, fresh walk stream); torus 300 epochs at lr 1e-4, fixed before launch (about the rate at which the source solution was reached).

A3. Batches

"Code check" = an independent, results-blind code review, whose findings were amended into the pre-registration before any result was read (before launch, except TEXTWORLD and LEAK, checked while the batch trained). "Reproduced" = the analysis was re-run from stored checkpoints by an independent reproduction audit (`docs/paper/review_draft5/REPRO_AUDIT.md`, `REPRO_AUDIT_nonsign.md`, 2026-10-16) and matched (spot re-evaluations within 7e-4); batches that landed after that audit read "no". "Attention": standard, or `--fast-attn` (SDPA; agrees to ~1e-6 in logits, not bit-identical in training). Pilots ran on seeds outside the batch except where noted.

batch	**main text**	**task**	**arms**	**seeds**	**epochs**	**attention**	**code check**	**reproduced**
PAPER2X2	5, 7.4	torus T = 128	RoPE, PoPE, MapWM r2 / r4, MapPoPE r2 / r4	0-7	300	standard	no	yes
SIGN_MATCHED		torus T = 1024		0-7	900	standard	no (authors'	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
	5, 7.3, 9		signed, |.|, softplus, RoPE				pre-launch gate)	
GENERALITY	5, 7.3	torus; ring p_plus 0.5	obs.-step-free signed / |.| and constructions; signed, CARoPE, CoPE, RoPE	700-707	900 (ring 300); +300	fast-attn (torus)	yes	yes
FACTORIAL	6	torus T = 1024	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	900	fast-attn	yes	yes
DYCK_MDEPTH	6	Dyck-2 L32 D12	MapWM, MapPoPE (r2), RoPE, PoPE at 1-4 layers; 3x; mixture	0-7	1x / 3x	standard	no	yes
CANCEL	6	ring p_plus 0.5-1.0	path 1 layer; index 1-3 layers	0-7; main text 2-7 (0-1 were the pilot)	300	standard	no	yes
RANK_MI	7.2	torus	shared r2, per-head r2, shared r4	0-7 (per-head r2 s0-1 were a pilot)	900	standard	no	yes
RANK_SEP	7.2	torus	block-diagonal r4, per-head r4	0-7	900	standard	no	yes
RANK3	7.2	torus	per-head r3	0-7	900	standard	no	yes
LOOP_RANK_E1800 (part 1)	7.2	torus	shared r2, shared r4	0-7	1800	standard	no	yes
RANK_PROJ	7.2	torus	projected r2 (frozen, trained), r4 control, r2 continued	0-7	+900	standard	no	yes
RANK_ND	7.2	2D (32 per side), 3D (10 per side) tori	per-head rank D, D + 1	0-7	900	standard	yes	yes
RANK_WRAP	7.2	2D and 3D tori of two sizes	per-head rank D + 1	0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
RANK_TESTS	7.2	ring; torus escape test	rank 1, 2 (+ variants); lifted / fresh rank 4	20-27; 0-7	300; 600	standard	yes	yes
SCORE_RANK	7.2, 7.4	torus T = 1024	MapWM / PoPE score at r2 and r4	r2 10-21, r4 10-17	900	standard	yes	no
LOOP_RANK	6, 7.4	torus T = 1024	r2 1 layer, r2 looped x4, r2 4 layers, r4	0-7 (A and C reused; A s0 reproduced bitwise)	900	standard	no	yes
RING_DIAL	7.3	ring p_plus 0.5 / 0.75 / 0.9	signed, |.|	30-37	300	standard	yes	yes
WRAP_EXISTENCE	7.3	ring p_plus 0.5	obs.-step-free signed / |.|, constructions	600-607	300 + 300	standard	yes	yes
LRSWEEP	7.3	ring	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2 / 3e-2	1500-1507	300	standard	yes	no (windowed run classes cross-checked)
ROBUST	7.3	ring (four recipes); torus 2 layers d 256	obs.-step-free signed / |.|	1400-1407	300; 900	fast-attn (torus)	yes	yes
ACCOUNTS	7.3	ring; line	T1 continuations; T2 frozen latents; T4 CoPE on a line	600-607; 1000-1007; 1100-1115	300	standard	yes	yes
MECHANISM	7.3	ring p_plus 0.5	eight start / step arms	800-807	300	standard	yes	yes
SIGNROUTE	7.3	ring, moves +1/-1	obs.-step-free signed, |.|, hybrid, abs_tied, exp, softplus at lr 1e-3 and 3e-3	1800-1807	300	standard	yes	no
SROPE_TORUS	7.3	torus T = 1024	Selective RoPE (best LM config), without SiLU, state-tracking	1700-1707	900	fast-attn	yes	no

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
			config, generator swap; MapWM; RoPE					
DELTA_FREE	7.4	torus T = 128, r2	PoPE / MapPoPE with delta pinned, free, frozen-random (path)	1600-1607	300	standard	yes	no
TEXTWORLD	8	torus in words	path 1 layer; index 1, 2 layers	0-7; main text 2-7 (0-1 were the pilot)	900	standard	yes	yes
TW_NORMSTEP	8	torus in words	MapWM, NormStep, DirOnly (direction words only step)	10-17	900	standard	yes	yes
LEAK	8	new-object torus	MapWM, ActOnly, NormStep	0-7	900	standard	yes	yes
GAIN_PHASE (K)	--	new-object torus	step x score 2x2	8-15	900	standard	yes	no
RING_BUDGET (H)	--	ring p_plus 0.5	|.| at 300 and 900 epochs	40-47	300 / 900	standard	yes	yes
TORUS_START (H)	--	torus	start arms	1200-1207	900	fast-attn	yes	yes
GAUGE (H)	--	ring, moves +1/-1 or +1/+2	obs.-step-free signed / |.|, RoPE	1300-1307	300	standard	yes	yes
WRAP_COPE (I)	--	ring	CoPE table starts	900-907	300	standard	yes	yes
MAPPOPE_PAIR (L)	--	torus T = 128	MapWM r2, MapPoPE r2 (32 / 64 angles)	10-25	300	standard	yes	no
RECENCY_EM, WARM, NOLEAK, SPREAD, SEARCH (N)	9	k-back recency	MapEM variants and installs	0-7	300 (SPREAD 1200)	standard	no	no
KNOB_SWEEP (A5)	9	torus T = 128, 16-	MapWM, RoPE under action variants	8 seeds	16	standard	not registered	no

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		epoch recipe						

ACCOUNTS T1's weight-decay-0.05 arms are bit-identical retrains of WRAP_EXISTENCE's constructed arms (a second readout of the same runs, not a replication). The shared rank-2 arm of every rank contrast is one stored set of eight runs. SIGN_MATCHED's signed arm is the source of the T = 1024 what/where analysis (App. K).

A4. Reproduction of MapFormer v4 Table 2 (2D)

With the paper's recipe (16 epochs x 98 batches, linear decay, lr 3e-4; `python3 -m mapformer.main --device cuda --epochs 16 --n-batches 98`), our MapWM scores 0.989 +/- 0.010 and our MapEM (single p_0) 0.987 +/- 0.012 on fresh maps at T = 128 (`PAPER_TASK_ACCURACY.md`), against 1.00 / 1.00 in v4 and 0.99 / 1.0 in v1 (MapWM-r2, MapEM-os, 2D IID).

A5. Egocentric actions (Sec. 9; `KNOB_SWEEP_n8.md`, not registered)

Torus at T = 128 with MapFormer's 16-epoch recipe, n = 8, held-out map. With allocentric moves (baseline) path integration's position effect over index RoPE is +0.438 (MapWM 0.967, RoPE 0.529). With egocentric turn-and-forward actions ("rotate") it is +0.050 (0.558 vs 0.508): a cumulative phase cannot integrate a heading-dependent step. Recoding the same walks into allocentric moves ("allocentric") restores it to +0.488 (0.996 vs 0.508). The index arm sits at the 0.508 floor in both action conditions, so these are boundary statements about the path model, not exchange rates.

B. Floors

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Rows marked (analysis) were computed after the batches.

task	predictor	value	stream	source
torus T = 128 (analysis)	blank / n-gram (orders 0-5) / retrace	0.507 / 0.598 / 0.934 (retrace per seed 0.928-0.946); wrap-only share 0.000	PAPER2X2 eval walks, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
torus T = 1024 (analysis)	blank / n-gram / retrace	0.506 / 0.575 / 0.843 (retrace per seed 0.837-0.849); wrap-only share 0.079	SIGN_MATCHED eval walks, seeds 0-7 (shared by RANK_MI, RANK_SEP, RANK3)	`docs/audits/2026-10-16/sign_floors_out.txt`
torus T = 1024 (analysis)	same	0.508 / 0.578 / 0.843; wrap-only 0.081	GENERALITY, seeds 700-707	`generality_floors_out.txt` (same directory)

task	predictor	value	stream	source
torus T = 1024, 2 layers d 256 (analysis)	same	0.509 / 0.578 / 0.844; every run of both arms above retrace (lowest |.| 0.910)	ROBUST, seeds 1400-1407	`robust_floors_out.txt`
torus T = 1024 (analysis)	blank / retrace	0.509 / 0.844	FACTORIAL, seeds 500-503	`docs/paper/figs/fig_factorial_strata_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 (analysis)	retrace	0.860 / 0.758 / 0.612 (blank 0.497-0.499; 0.864 at p_plus 0.5 on the registered eval stream)	2000 walks per bias, another stream of the same task	`docs/audits/2026-10-16/ring_floors_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 / 1.0	better of best constant and n-gram (orders 1-5)	0.554 / 0.544 / 0.520 / 0.498	1000 held-out walks	`CANCEL_PREREG.md`, `CANCEL_RESULTS.md`
line (256 cells)	blank	0.503	ACCOUNTS T4	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	blank (retrace reduces to it)	~0.50	GAUGE	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy	0.505 / 0.597	TEXTWORLD eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best stack-free A2f floor (n-grams of orders 1-6)	0.594	gate G4	`DYCK_MDEPTH_RESULTS.md`

Per-stratum floors (analysis). Torus, seeds 0-7 (`docs/paper/figs/torus_strata_floors_s0-7.json`): best constant 0.506 / 0.500 / 0.507 (short gap / long gap / wrap-only), retrace 0.911 on short gaps and equal to the constant on the others (walk runs are at most 10 moves, so a retrace never reaches a gap of 128 moves or a wrap-only revisit). FACTORIAL's stream: 0.508 / 0.515 / 0.517, retrace 0.912. Ring (`ring_strata_floors_out.txt`): retrace 0.9998 at gaps 2-7, 0.943 at 8-31, 0.570 at 32-127, 0.493 wrap-only; constant 0.49-0.50 in every stratum.

Against its own seed's retrace floor (analysis, torus T = 1024): signed falls below on 0/8 runs, |.| on 3/8 (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8.

Further floors (analysis unless marked).

task	predictor	value	stream	source
torus T = 1024				`SCORE_RANK_RESULTS.md`

task	predictor	value	stream	source
	blank / n-gram / retrace	0.507 / 0.576 / 0.843	SCORE_RANK (registered in its results file)	
torus T = 1024	blank / n-gram / retrace	0.509 / 0.578 / 0.845	SROPE_TORUS, seeds 1700-1707	`SROPE_TORUS_RESULTS.md`; `docs/audits/2026-10-20/srope_torus_floors_out.txt`
torus T = 128	retrace	0.938	DELTA_FREE, seeds 1600-1607	`DELTA_FREE_RESULTS.md`; `docs/audits/2026-10-20/delta_free_floors_out.txt`
torus in words	always-'nothing' / one-step reversal-copy / retrace	0.5054 / 0.5965 / 0.9337	TEXTWORLD registered eval stream	`docs/paper/expansion/scripts/language/tw_retrace_floor_out.txt`
torus in words	same	0.5120 / 0.6024 / 0.9365	TW_NORMSTEP eval stream	same
torus in words, revisits retrace cannot reach	best constant	0.506 ('nothing'; 856 of 6,381 revisits, 13.4%)	all 200 walks of the TEXTWORLD stream	`docs/paper/expansion/scripts/language/tw_strata_200walks_out.txt`
torus in words, ... whose cell holds an object (the revisits retrace gets wrong)	best constant ('nothing' excluded by construction)	0.088 ('cat'; 423 of 6,381, 6.6%)	same	same
ring, moves +1/-1, observation-step-free	n-gram / retrace	0.554 / 0.860	SIGNROUTE reference	`SIGNROUTE_RESULTS.md`

C. Per-seed tables

Accuracy at the training length on held-out maps, eval mode. SOLV / STAL / DESC = SOLVED / STALLED / DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Secs. 5, 7.3; `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)

seed	signed	|.|	softplus	index RoPE
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Registered: |.| - signed -0.177 (permutation p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014); signed - RoPE +0.267 (p 0.0002; 8/8 vs 0/8, p 0.0002). r(final loss, accuracy) -0.992 over 32 runs.

C2. Torus, index depth (Sec. 6; `FACTORIAL_{L1,L4,LOOP}.json`)

seed	path, 1 layer	index, 1 layer	index, 4 layers	index, looped x4
500	1.000	0.719	0.814	0.868
501	1.000	0.731	0.870	0.861
502	1.000	0.688	0.874	0.860
503	1.000	0.739	0.881	0.868

Every path run SOLVED, every index run STALLED. Registered C0-C3: path beats index at one layer (+0.281), four layers (+0.141) and loop x4 (+0.136); depth and loop each help index (+0.140, +0.145); every firing a complete separation (p 0.029).

C3. Ring exchange rate, fresh seeds 2-7 (Sec. 6; `docs/audits/2026-09-27/fresh_seeds.txt`)

p_plus	path 1 layer	index 1 layer	gap to path at 1 / 2 / 3 index layers
0.5	1.000	0.710 +/- 0.020	+0.290 / +0.051 / +0.002
0.75	1.000	0.677 +/- 0.009	+0.323 / +0.038 / +0.002
0.9	1.000	0.824 +/- 0.007	+0.176 / +0.013 / +0.001
1.0	1.000	1.000	0 / 0 / 0

All eight seeds (`CANCEL_RESULTS.md`): index 1 / 2 / 3 layers 0.717 / 0.954 / 0.998 (p_plus 0.5), 0.676 / 0.969 / 0.998 (0.75), 0.825 / 0.988 / 0.999 (0.9), 1.000 (1.0); path 1.000 in every cell (32/32 SOLVED); registered exchange rate k = 3, 3, 3, 1 index layers.

C4. Rank (Sec. 7.2; `RANK_MI_ANALYSIS.txt`, `RANK_SEP_ANALYSIS.txt`, `RANK3_ANALYSIS.txt`, `RANK_SEP.json`, `RANK3.json`)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
0	0.849 STAL (0.587)	0.656 STAL (1.049)	0.879 STAL (0.482)	0.942 DESC (0.243)	1.000 SOLV (0.004)	0.996 SOLV (0.004)
1	0.998 DESC (0.051)	1.000 SOLV (0.007)	0.936 DESC (0.302)	0.990 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.010)
2	0.917 DESC (0.308)	1.000 SOLV (0.003)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
3	0.917 DESC (0.321)	0.723 STAL (0.910)	0.945 DESC (0.234)	0.967 DESC (0.134)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
4	0.947 DESC (0.177)	0.959 DESC (0.194)	0.964 DESC (0.202)	1.000 SOLV (0.003)	1.000 SOLV (0.002)	0.998 SOLV (0.002)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
5	0.907 STAL (0.404)	0.949 DESC (0.211)	0.917 DESC (0.310)	1.000 SOLV (0.006)	0.994 SOLV (0.004)	1.000 SOLV (0.008)
6	0.773 STAL (0.675)	0.943 DESC (0.248)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.004)	1.000 SOLV (0.005)
7	0.842 STAL (0.605)	0.851 STAL (0.539)	0.947 DESC (0.166)	1.000 SOLV (0.005)	0.987 SOLV (0.010)	1.000 SOLV (0.003)
mean	0.894, 0/8	0.885, 2/8	0.948, 2/8	0.987, 6/8	0.998, 8/8	0.999, 8/8

Per-head r2 seeds 0-1 were a pilot run before registration (1/6 SOLVED on seeds 2-7). Registered: per-head rank 4 vs block-diagonal 8/8 vs 2/8 (Fisher and permutation p 0.0070); shared r4 vs block-diagonal the same (it also changes per-head rank); sharing (per-head r4 vs shared r4, +0.001) and W_out scale (block-diagonal vs per-head r2, +0.063) not detected; per-head r3 vs r2 +0.102 (permutation p 0.027; SOLVED 6/8 vs 2/8, Fisher p 0.13); r4 vs r3 +0.012, not detected. Shared r4 vs shared r2: 8/8 vs 0/8 (Fisher p 0.0002), +0.104 (p 0.0005).

Twice the budget (`LOOP_RANK_E1800_P1_ANALYSIS.txt`, 1800 epochs from scratch): shared r2 0/8 (final-5% losses 0.489, 0.383, 0.390, 0.340, 0.141, 0.231, 0.305, 0.571; 3 STALLED, 5 DESCENDING), accuracy 0.908; shared r4 7/8 (s5 STALLED at 0.180), 0.994.

C5. Sign, ring and torus (Sec. 7.3)

Ring dial (`RING_DIAL_VERDICTS.json`; full models, S = SOLVED):

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Registered: p_plus 0.5 -0.028 (permutation p 0.007), 2/8 vs 8/8 (Fisher p 0.0070); 0.75 -0.032 (p 0.0002), 1/8 vs 8/8 (p 0.0014); at 0.9 the contrast did not reach the registered threshold (-0.046, below the 0.05 materiality; 4/8 vs 8/8, p 0.077).

Observation-step-free ring from scratch and constructed (`WRAP_EXISTENCE_ANALYSIS.txt`):

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)
602			1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
	1.000 S	0.853 STAL		
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Registered: from scratch 0.921 vs 1.000, -0.079 (permutation p 0.0002), 0/8 vs 8/8 (Fisher p 0.0002); constructed |.| HELD (8/8 SOLVED, 0/8 LOST).

Observation-step-free torus (`GENERALITY_ANALYSIS.txt`):

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139

Registered: from scratch 0.880 vs 1.000, -0.120 (p 0.0002), 0/8 vs 8/8; the constructed |.| model leaves the solved region on 6/8 seeds (peak loss 0.11-0.21; signed control 0.004-0.017) and ends SOLVED on 8/8 at 1.000: by the reading fixed before launch, perturbed and re-found.

Recipe and scale (`ROBUST_ANALYSIS.txt`, seeds 1400-1407; accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
```

```
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

Learning-rate sweep (`LRSWEEP_RESULTS.md`, seeds 1500-1507): per-seed table in H2.

C6. Text world (Sec. 8; `TEXTWORLD_ANALYSIS.txt`)

seed	path, 1 layer	index, 1 layer	index, 2 layers
0 (pilot)	0.881	0.504	0.695
1 (pilot)	0.998	0.504	0.723
2	0.999	0.503	0.927
3	0.995	0.506	0.722
4	0.999	0.505	0.926
5	1.000	0.506	0.706
6	1.000	0.506	0.760
7	0.882	0.504	0.720

Fresh seeds 2-7 (`docs/audits/2026-09-27/fresh_seeds.txt`): path 0.979 +/- 0.048 (6/6 SOLVED), index 1 layer 0.505 (0/6), index 2 layers 0.793 (0/6); path - index one layer +0.474, permutation and Fisher p 0.0022. All eight seeds: path 0.969 +/- 0.054 (7/8; seed 0 still descending), +0.464 (p 0.0002; Fisher p 0.0014); path - index two layers +0.197 (p 0.0009).

C7. Dyck-2

Per-seed values are in `DYCK_MDEPTH_RESULTS.json` (cells `T12|{RoPE,PoPE,MapWM,MapPoPE}|{1-4}L|L32D12`); every one of the eight seeds has a positive position effect at every depth (sign-flip p 0.0078 per depth).

C8. SOLVED-threshold sensitivity

Sign batches (`docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`; predictions and decision rules written before the re-analysis; two-sided Fisher p in brackets):

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	4/8 vs 8/8 (0.0769)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
WRAP_EXISTENCE ring	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	|.| vs signed, same-direction start	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
ACCOUNTS T2 ring	frozen vs free move latents	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
GENERALITY torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)

ROBUST (declared secondary, loss < 0.1): ring default 1/8 vs 8/8 (p 0.0014), weight decay 0 1/8 vs 8/8 (0.0014), lr / 3 0/8 vs 7/8 (0.0014), lr x3 4/8 vs 8/8 (0.077), torus 2 layers d 256 5/8 vs 8/8 (0.20); on that torus cell the accuracy contrast (0.967 vs 0.999, p 0.0005) carries the result. LRSWEEP (loss < 0.1, selection redone): 2/8 vs 8/8, same label.

Rank batches (analysis; read from the per-seed final-5% losses in C4): at loss < 0.02 the counts are 0 / 2 / 2 / 6 / 8 / 8 of 8 (shared r2, per-head r2, block-diagonal r4, per-head r3, shared r4, per-head r4), identical to the registered 0.05; at loss < 0.1 they are 1 / 2 / 2 / 6 / 8 / 8 (shared r2 seed 1 at 0.051 enters).

C9. Attention-dropout evaluation scale

Models trained with attention dropout expect the 1/(1 - p) attention scale; evaluated with it, below-ceiling one-layer runs score higher. The registered text-world evaluation re-run with the scale (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`, all 24 runs) gives path 0.969 -> 0.993, index one layer 0.505 -> 0.506, index two layers 0.772 -> 0.772; path - index one layer +0.464 -> +0.488 (p 0.0002). The re-score was a declared secondary of RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE and changed no label there; it was applied to SIGN_MATCHED by the same audit with no verdict changed. ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (the correction does not apply at two layers). Main-text accuracies are eval mode without the scale except where marked re-scored (Sec. 8, Table 7, App. J).

C10. Sign route (Sec. 7.3; `SIGNROUTE_RESULTS.md`, registered; ring, observation-step-free, seeds 1800-1807)

Accuracy at T = 128 and run class (S SOLVED, T STALLED, D DESCENDING; none diverged):

```
lr     arm       1800   1801   1802   1803   1804   1805   1806   1807
1e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs       0.912T 0.874T 0.902T 0.926T 0.918T 0.968T 0.939T 0.902T
1e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs_tied  0.748T 0.748T 0.672T 0.723T 0.749T 0.751T 0.748T 0.770T
1e-3   exp       0.904T 0.973T 0.929T 0.957T 0.852T 0.958T 0.936D 0.972S
1e-3   softplus  0.841T 0.794T 1.000S 0.984T 0.734T 0.861T 0.909T 0.906D
3e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs       0.921T 0.944T 0.935T 0.968T 0.949T 1.000S 0.945T 1.000S
3e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs_tied  0.743T 0.748T 0.652T 0.778T 0.775T 0.748T 0.774T 0.769T
3e-3   exp       0.897T 0.959D 1.000S 1.000D 0.902T 0.967D 0.960D 0.977D
3e-3   softplus  0.954T 0.959D 0.999S 1.000S 0.964T 0.975D 1.000S 0.880T
```

D. Constructions

D1. The rank-2 projection (Sec. 7.2; `RANK_PROJ_RESULTS.md`)

Each solved shared rank-4 model (seeds 0-7, a stored batch) is projected onto the top two directions of its action latent over the vocabulary (top-2 spectral energy 0.99952-0.99994), every other weight kept, and installed in the shared rank-2 architecture. Frozen, the projections score 0.9955 at T = 1024 against their sources' 0.9966 (at least 0.981 on every seed; per seed 0.9993, 1.0000, 0.9988, 1.0000, 0.9850, 0.9806, 1.0000, 1.0000). Trained 900 epochs from a learning-rate restart at 1e-3, 7/8 are SOLVED (one DESCENDING), as for the rank-4 control trained from its own solution (7/8); rank 2 continued from its own 900-epoch weights solves 1/8. Registered verdict: STABLE.

D2. Per channel (Lemma P1.0 in v3)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The model reads each channel's angle only through cos and sin, so the model with Delta' is the same function.

D3. The balanced |.| construction (Proposition 3(a), E2; the registered existence test)

Given a solved signed observation-step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4):

1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings, so this is a free gauge.
2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega).

3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere; copy every other weight.

The signed control is built by the same function with the unwrapped signed angle (steps omega Delta / omega'), so the two differ only in the wrap. On the torus m = 4 = r, exactly at Proposition 3(a)'s limit.

Exactness. Ring: frozen accuracy = source on 8/8; maximum angle error 1.2e-6 rad (|.|), 6.1e-7 (signed); maximum logit difference 5.1e-4 at T = 128. Torus: frozen = source on 8/8; maximum angle error 1.4e-6 rad (|.|), 5.8e-7 (signed). *Weight scale.* Ring: maximum |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built; |W_in| 0.15-0.21 -> 0.19-0.33. Torus: maximum |W_out| 0.48-0.58 -> 1.23-1.55, maximum |omega| 0.68-0.96 -> 1.00. Within about 5x of the trained scale.

Why the scale matters. A construction that keeps the source's W_in and solves W_out at minimum norm is also exact, but in a solved signed model the two move latents are nearly antiparallel (cosine -0.987 to -0.998), so W_out must be huge (up to 16,320 against 0.31-0.38). Adam's per-entry steps then lose that solution in the warm-up (8/8 LOST) before re-finding it (7/8 SOLVED). An existence test is informative only near a trained scale, which the balanced construction provides.

D4. The full |.| arms

To show that the task, not only a given signed table, is representable by a full |.| model (observations stepping), W_in is chosen to project onto two directions orthogonal to all 17 observation embeddings and W_out is solved per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1). In the real modules: |.| steps >= 0, observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs.

D5. Softplus (Proposition 3(f), E2)

Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi. With W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) / omega'). Built in the real softplus model from a solved ring model (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): minimum step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the same construction applies.

D6. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. *LOST* (registered): the run's maximum per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's maximum at that seed and 0.05. *HELD* = SOLVED on at least n - 1 seeds, mean accuracy at least control - 0.01, and LOST on at most one seed. Ring (lr 1e-3): HELD, 8/8 SOLVED, 0/8 LOST (maximum per-epoch loss 0.024-0.154 against the control's 0.015-0.112). Torus (lr

1e-4): LOST on 6/8 and SOLVED on 8/8 at the end, read by the rule fixed before launch as perturbed and re-found; at lr 1e-3 (descriptive) 6/8 SOLVED (0.988).

D7. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (mean query norm at move tokens times mean key norm at observation tokens, per channel), the held ring |.| model keeps its cancelling code from 0.99 at construction to 0.98 after training; unweighted (every channel equal) the share falls from 0.87 to 0.18, because channels attention does not read leave the code. Weight decay 0 vs 0.05 changes nothing detected (-0.003).

E. Theory

Labels used here: **proved** (a proof is given), **cited** (a published construction applied to our setting), **conjecture** (a derivation whose assumptions or dynamics are not checked). Main-text numbering: Proposition 1 = E6.1; Proposition 2 = E7.1 + E7.2; Proposition 3 = E2 (a), (f), (g); Lemma 1 = E3; Proposition 4 = E8.1-E8.6; Proposition 5 and Conjecture 1 = E9.2-E9.3; Remark 2 = E10; Proposition 6 = E11; Proposition 7 = E12; Proposition 8 = E5; Remark 3 = E13; Remark 1 = E6.2. The check scripts named "(check)" are CPU-only and eval-only; those in `docs/paper/appendix_checks/` were refereed with the v3 theory appendix, those in `docs/paper/expansion/scripts/` are listed in App. O.

Notation. A rotary channel is c = (h, b), head h and block b, acting on query and key coordinates (2b, 2b + 1) of head h; R(phi) is the 2D rotation. The phase is theta_{t,c} = omega_c sum_{u <= t} S_{c, x_u}, with S = W_out W_in E the C x V step table (rank at most r) and Theta = diag(omega) S the angle table; omega is free in sign. For token x the latent is z_x = W_in e_x and channel c's readout row is w_c, so the pre-nonlinearity step is w_c . z_x and the angle a_c(x) = omega_c w_c . z_x (signed), omega_c |w_c . z_x| (|.|), omega_c softplus(w_c . z_x) or omega_c exp(w_c . z_x). For an inverse pair of tokens (x, y), such as opposite moves, write the clock k_c = a_c(x) + a_c(y), the map m_c = a_c(y) and delta_c = (a_c(y) - a_c(x)) / 2. Channel c *cancels through zero* if k_c = 0 with a_c(x) ! = 0, *cancels by wrapping* if k_c is a non-zero multiple of 2 pi, and is *blind* if both angles are 0 mod 2 pi. *Usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens (exact per token at layer 0).

E1. Score identity (proved; the identity is Mamba-3's Proposition 3)

With alpha piecewise constant, alpha = Delta_u on (u - 1, u], the homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has propagator F(t) = exp(-s (Phi_t - Phi_{t'})) F(t'). With s = i omega, channel c of the rotary score is q_c^T R(theta_s - theta_t) k_c = Re(conj(q~*c) k~_c exp(-i omega_c (Phi_t - Phi_s))), with q~ = q*{2b} + i q_{2b+1}.

Proof. On each interval the ODE is linear with constant coefficient -s Delta_u; flows over tokens compose by adding exponents. R(theta) acts on q~ as multiplication by e^{i theta}, so (R(theta_t) q)^T (R(theta_s) k) = Re(conj(q~) k~ e^{-i (theta_t - theta_s)}). [] With input, the zero-order-hold solution is F(t) = e^{-s Phi_t} F(0) + sum_{u <= t} e^{-s (Phi_t - Phi_u)} ((1 - e^{-s Delta_u}) / s) f_u; a zero-rate token writes nothing there, whereas in attention a zero-step token still writes its key and value at full strength. *Check*

(`a0_propagator.py`; `review_draft5/theory_check_extra.py` (1)): 3e-14 (propagator), 4e-15 (score), 1.1e-14 (input term).

E2. Representability by wrapping (proved)

(a) At most r stepping tokens. Let S be a signed table with at most m <= r non-zero columns (stepping tokens x_1, ..., x_m), whose embeddings are linearly independent modulo the span of the other embeddings (generic when V <= d_model, here 19 or 21 <= 128; automatic in observation-step-free models). Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j (the j-th unit vector of R^r) and W_in' e_y = 0 for every other token. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0, other columns zero. Then W_out' W_in' E is entrywise non-negative, so |.| leaves it unchanged, and it reproduces each stepping column mod 2 pi. [] *Full arms:* choosing W_in' to annihilate every observation embedding leaves only the moves stepping, so every |.| model in the paper can represent any code in which only the moves step. *Check* (`p1_wrap.py`): angle error <= 7e-14, logits <= 8e-14.

(f) Softplus. Under the same assumptions, send the stepping embeddings to u_1, ..., u_m and every other embedding to 0, set omega'*c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi. *Proof.* softplus is a bijection from R onto (0, inf); the targets lie in [ln 2, 2 ln 2); omega' ln 2 = 2 pi. [] *Check* (`theory_check_extra.py` (2)): built in the real softplus model, logits equal to the source's to 2.8e-11.

(g) Exp. The same construction with omega'*c = 2 pi and W_out'[c, j] = ln((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / 2 pi) makes every non-stepping token step exp(0) = 1, an angle of exactly 2 pi, and every stepping token's angle Theta mod 2 pi. *Proof.* exp is a bijection from R onto (0, inf); the targets (remainder + 2 pi) / 2 pi lie in [1, 2). [] Not built in a trained model.

What representability does not cover. For a rank-1 |.| model and two stepping tokens with steps of strictly opposite sign, exact representation mod 2 pi fails for almost every signed table (v3 P1(c), proved); rank-1 non-negative codes approximate any table mod 2 pi at a large angle scale (P1(d), Kronecker). Every claim in the paper uses (a), (f) or the full-arm construction, which are exact.

E3. The relative-sign gauge (Lemma 1; proved)

Take a MapWM model with any number of layers and a set F of channels. Form a second model by (a) negating every token's step in each channel of F (by negating omega_c; in the signed class equivalently row c of W_out, which is not available in the |.| class since |.| is applied after W_out) and (b) negating output coordinate h d_head + 2b + 1 of the query and key projections (weight row and bias entry, every layer) for each c = (h, b) in F. The two models have identical attention scores at every layer, hence identical logits.

Proof. With J = diag(1, -1), J R(phi) J = R(-phi) and J is orthogonal. (a) replaces theta_{t,c} by -theta_{t,c}; (b) replaces q_c, k_c by J q_c, J k_c, exactly including biases. The channel term becomes (J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T R(theta_s - theta_t) k_c.

Every layer uses the same cos / sin, so the scores agree layer by layer. [] *Check* (`t1_gauge.py`): 35 channels flipped and reflected give bitwise-identical logits (control without the reflection: 6.6).

Corollaries. (i) The |.| class realises a signed angle table Theta exactly as real numbers, with the same latent, iff every row of Theta is sign-coherent. (ii) It therefore excludes no map *function* for which a code with sign-coherent rows exists, and that code needs no wrapping: one stepping token (parity, Z_m counting) or two moves that step the same way (+1 and +2: a(+2) = 2 a(+1)). (iii) Up to the gauge, the excluded tables are exactly those with a row containing entries of strictly opposite sign. For an inverse pair every through-zero cancelling channel is such a row; cancellation by wrapping, (a, 2 pi - a), is allowed whenever E2 applies. The constraint removes the through-zero branch of the inverse-pair problem, not cancellation.

E4. One channel's geometry (proved)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), everything else fixed; the loss is a function on the angle torus. Write sigma = a_x + a_y.

E4.1 (cut circle). (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed and |.| classes attain the same loss values in this channel. (b) The non-blind cancelling set in the quadrant consists of the segments sigma = 2 pi k, k >= 1; the through-zero line meets the quadrant only at the blind origin. (c) From a start in the open quadrant with sigma < 2 pi, the infimum distance (in angle coordinates) to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class; they coincide iff sigma >= pi. (d) Any continuous |.| path from |sigma| < pi to a non-blind cancelling point passes through |sigma| = pi.

Proof. (a), (b) from the quotient map. (c) The distance to sigma = 2 pi k is |sigma - 2 pi k| / sqrt 2; the |.| class has only k >= 1. (d) |sigma| is continuous along a |.| path and a non-blind cancelling point has |sigma| >= 2 pi. [] *Check* (`p3_p5_geometry.py`): a brute-force grid matches both formulas to 2e-4.

E4.2 (sticky wall). Let s be the pre-|.| step of one token in one channel and a = omega |s|, with L(s) = l(|s|) and l continuously differentiable near 0. (a) For s != 0, dL/ds = omega sign(s) dL/da. (b) If l'(0) > 0, s = 0 is a strict local minimum of L along s with every other parameter fixed. (c) Gradient descent on s with step eta, l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M). *Proof.* Chain rule; first-order expansion; case split. [] In parameter space s = w_c . z_x is shared across tokens and channels, so (b) is not a local minimum of the loss. In trained |.| models the wall is the cone boundary of E8.2 (App. H7).

E5. CARoPE cannot cancel (Proposition 8; proved)

CARoPE (faithful, `model_mono_baselines.MapFormerWM_CARoPEF`) advances channel i of head h by f_h(x)^i per token, with f in (0, 1), so every per-token advance lies in (0, 1]; channel 0 advances by exactly 1, an index clock. (a) The phase of any interval of k >= 1 tokens lies in (0, k]. The shortest return spans 3 tokens between the move query and the observation key, and 4 between two observation tokens, so its phase lies in (0, 4], a subset of (0, 2 pi): no channel, at any parameter value, reads every return at phase 0. (b) Fix eps in (0, pi -

2). If a channel reads each return of an excursion of J moves out and J back within eps of 0 mod 2 pi, then the move-driven phase of any interval with at most 2J moves lies in [0, 2 eps]: every channel is either a clock or blind over that range.

Proof. (a) Sum k numbers in (0, 1]. (b) The depth-j return phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j in (0, 4]; the arc (eps, 2 pi - eps) is longer than 4 when eps < pi - 2, so a sequence starting in (0, eps] with increments at most 4 cannot jump it; induction gives P_j in (0, eps], hence J (c_+ + c_-) <= eps. [] *Check* (`p6_carope.py`): eight trained ring CARoPE models; 10,331 random draws satisfy the bound.

E6. Path integration supplies the prefix sum; index attention needs a count

E6.1 (Proposition 1; proved). Tokens interleave moves and observations, [a_1, o_1, a_2, o_2, ...]; o_t is predicted at the token a_t. Let P_t be the unwrapped displacement after move t. The map is redrawn per evaluation, so the only source of o_t is an earlier observation o_s with P_s = P_t (mod N per axis). In MapWM, with observation steps zero and a fixed step vector per move, theta_t = omega P_t and score(t, s) = q^T R(omega (P_t - P_s)) k: the predicate is computed outside attention, exactly (a float32 cumsum), for every gap and every T. In a channel whose frequency times the move step is a multiple of 2 pi / N, the score is N-periodic in P, and wrap-around costs nothing. [] In an index-RoPE layer score(t, s) = q(h_t)^T R((t - s) Omega) k(h_s): position enters only through t - s, so the predicate "the moves in (s, t] sum to zero" must be written into the residual stream h.

E6.2 (Remark 1; cited). *Route A, a global count.* Layer 1: one head with constant query and key attends uniformly to the prefix; its value carries the move vector on move tokens, so its output is P_t / n_t. A feature proportional to 1/n_t is available in the same layer (for example a head with a fixed self-attention bonus). The MLP rescales to P_t and forms periodic features cos(2 pi k P_t / N), sin(2 pi k P_t / N); layer 2 matches on them. This is Liu et al.'s (2023) Lemma 6 followed by a match, and Yao et al.'s (2021) depth-2 Dyck construction (as summarised in Liu et al., App. A.5). In RASP terms the task has depth 2 in 1D and 2D alike. Its costs: a normaliser (Lemma 6 uses a padding token and an absolute channel log(2T - t); our torus and ring sequences have neither, Dyck has a BOS token); precision 1/n_t (1/256 at the end of a ring walk, 1/2048 per axis at the end of a torus walk); MLP width 4nT and weights up to 4nT + 2 in Lemma 6's mod gate (32 x 256 for the ring, 64 x 2048 per axis for the torus; our MLP has width 512); and sensitivity growing with the gap (Hahn and Rofin 2024, Cor. 5 and Thm. 6, Boolean inputs; used for direction only). *Route B, local return rules.* Retrace (while the current run reverses the previous one, the cell is the one seen 2j moves, 4j tokens, earlier: a bounded-window rule) and the one-lap clock (when the walk never reverses, the cell N moves back) need no count. A return that spans k direction runs is a function of those k runs; selecting its offset needs a summary of k runs, and stacking layers extends the reach. The reach is measured (App. F), not derived.

E7. Clock, map, the position-factor bound and the tick rule (proved)

E7.0 (phase decomposition). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase of channel c is a_c(y) n_y + a_c(x) n_x = m_c d + k_c n_x, exactly.

E7.1 (Proposition 2(a), a bound on the position factor). Let a revisit at displacement 0 have n backward moves; its phase in channel c is k_c n. For usage weights q_c (sum 1), the same-cell position factor F(n) = sum_c q_c cos(k_c n) equals 1 at n = 0; if a usage share s of the channels has kappa <= |k_c| <= K (unwrapped), then F(n) <= 1 - s (1 - cos(kappa n)) for n <= pi / K.

Proof. phi = m d + k n with d = 0. For n <= pi / K, kappa n <= |k_c| n <= pi, and cos is decreasing on [0, pi], so cos(k_c n) <= cos(kappa n) on the share s and <= 1 elsewhere. [] The bound covers n <= pi / K only. It says how fast a clock erodes the position factor of a revisit; it is not a theorem that a clocked code loses long-gap revisits, and near |k_c| n = pi it says nothing: two signed torus models started past pi solve with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi (App. H7). Index RoPE is the extreme case: every token, observations included, advances every channel, so the whole code is clock. Wrap-only revisits also need the lattice condition N m_c = 0 mod 2 pi.

E7.1' (the measured clock / map ratio). Every clock / map number in the paper (Table 4, G3, H7) is rho_cm = sqrt(sum_{pairs} sum_c q_c wrap(k_c)^2 / sum_{pairs} sum_c q_c (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2), with the usage q normalised over the channels of both heads (one value per run), or over one head's channels (per head). The symmetric denominator makes it invariant to which move of a pair is called y (rule 8); for unwrapped |.| channels |k_c| >= |m_c| (E8.1), so rho_cm >= 1 there by construction. Generator: `scripts/sign/clock_table.py`.

E7.2 (Proposition 2(b), tick rule). Split each move's contribution in channel c into the map part and a tick tau_c, the phase every move adds whatever its direction. If the target at t is the key at the last visit to the same cell, matched pairs come at lags L drawn from a distribution, and with the map part cancelling over the closed loop the matched phase difference is L tau_c. A channel scores matched pairs consistently only if L tau_c is constant mod 2 pi over the lags that occur, i.e. tau_c = 0 mod 2 pi / g, g the greatest common divisor of the return lags (g = 2 on a bipartite lattice such as the even torus). Otherwise its matched score decorrelates once |wrap(tau_c)| sd(L) exceeds about 1 rad. [] A pi tick is therefore invisible to revisits on the even torus: it encodes the parity of the cell.

E8. Route inventory (Proposition 4; proved)

E8.1 (the |.| clock floor). In the |.| class, for an inverse pair in channel c (one omega_c of either sign): (a) a_c(x) and a_c(y) have the same sign, so |k_c| = |a_c(x)| + |a_c(y)| >= max(|m_c|, 2 |delta_c|), with |k_c| = |m_c| iff a_c(x) = 0 and |k_c| = 2 |delta_c| iff one of the two angles is 0. (b) As real numbers k_c = 0 iff the channel is blind, so the non-blind zero-clock points are exactly the wrap sheets |k_c| in 2 pi Z_{>=1}. (c) On a zero-displacement revisit with n = g/2 backward moves the phase is k_c g / 2 >= |m_c| g / 2; if |k_c| < 2 pi it is within eps of 0 mod 2 pi only for g <= 2 eps / |k_c| <= 2 eps / |m_c| (or g near a multiple of 4 pi / |k_c|).

Proof. (a) same sign of omega_c times non-negative numbers; triangle equality cases. (b) from (a). (c) from E7.0. [] An unwrapped |.| channel is at least as much clock as map, and the better it resolves neighbouring cells, the shorter the backtracks it can read. E4.1 is the geometric form of (b). *Check* (`sign/sign_clock.py`): zero violations of the inequality in 4,059 sub-pi |.| channels.

E8.2 (the collective route of the signed class). Signed class, inverse pair, readout W in R^{C x r} of full column rank, omega_c != 0. (a) The clock vector is k = diag(omega) W u with u = z_x + z_y. The latent pairs at which every channel has zero clock form the r-dimensional linear subspace {u = 0}; the map part (z_y - z_x)/2 is free on it, so it contains non-blind points. (b) Along z_y(tau) = (1 - tau) z_y - tau z_x (W, omega fixed), u(tau) = (1 - tau) u(0), and every channel's clock is (1 - tau) k_c(0). (c) For the quadratic surrogate R(z) = (1/2) sum_c q_c k_c^2, gradient flow on (z_x, z_y) gives du/dt = -2 H u with H = W^T diag(omega) diag(q) diag(omega) W, positive definite when the used channels' rows span R^r, and the map part is constant: u(t) = exp(-2 H t) u(0) -> 0 from every start, and {u = 0} is the only stationary set. (d) For almost every (W, omega), {u = 0} is the only latent set at which every channel cancels mod 2 pi when C > r.

Proof. (a)-(c) linear algebra: k_c = omega_c (w_c . z_x + w_c . z_y) and dR/dz_x = dR/dz_y = H u. (d) diag(omega) W u is in 2 pi Z^C iff (2 pi n_c / omega_c)_c lies in col(W); for fixed n != 0 the map omega -> (2 pi n_c / omega_c) is a diffeomorphism of (R \ 0)^C onto itself, so the preimage of the r-dimensional subspace col(W) is null when r < C; take the countable union over n. [] The surrogate guarantee concerns the clock term alone; with the content frozen, gradient descent on the step map rarely takes this route from the same-direction start (App. H7).

E8.3 (no collective route for |.| on an inverse pair). |.| class, C channels, rank r. (a) The latent route of E8.2(b) ends, under |.|, at a(y) = a(x) in every channel, since |w . (-z_x)| = |w . z_x|: a clock. (b) For almost every (W, omega), at that fixed (W, omega) no latent pair (z_x, z_y) makes more than r channels of each sign-coherence class (pre-|.| steps of equal or of opposite sign), hence more than 2r channels, cancel non-blindly. (c) Hence every |.| map code with more than 2r cancelling channels has its readout tuned channel by channel: in step units the channel-c condition is |w_c . z_x| + |w_c . z_y| = 2 pi n_c / |omega_c|, an offset that differs across the frequency ladder (omega_max / omega_min = 32 on the ring, 64 on the torus).

Proof. (a) evenness of |.|. (b) Fix a set S of 2r + 1 channels, a sign pattern per channel and non-zero integers n_c (a non-blind |.| channel cancels only by wrapping, E8.1(b)). On the sign region k_c = omega_c (e_c(x) w_c . z_x + e_c(y) w_c . z_y) is linear in zeta = (z_x, z_y) in R^{2r}, k = diag(omega) N zeta, and the conditions k_c = 2 pi n_c for c in S hold for some zeta iff (2 pi n_c / omega_c)_{c in S} lies in col(N), a subspace of dimension <= 2r < 2r + 1 that does not depend on omega. As in E8.2(d) the set of omega_S for which it holds is null; Fubini and a countable union finish. Within one coherence class the sum depends on zeta only through z_x + z_y or z_x - z_y, an r-vector, giving the per-class bound. [] *Check* (`sign/prop_checks.py` (1)): at the default readouts of eight seeds, least-squares latents never cancel 9 channels exactly (smallest maximum error 0.027 rad); 8 channels cancel exactly (1e-14) only when they split 4 / 4 between the coherence classes.

E8.4 (same-direction moves). For a relation n_x a(x) + n_y a(y) = 0 (+1/-1: n = (1, 1); +1/+2: n = (-2, 1)): signed has one linear zero set; |.| with n_x n_y < 0 (no inverse pair) has two collective zero sets, {n_x z_x + n_y z_y = 0} and its mirror under z_y -> -z_y, both with non-blind points; with n_x n_y > 0 (an inverse pair) neither is non-blind. In the quadratic surrogate restricted to a sign region in which both coherence classes contain channels whose rows span R^r, the only latent stationary point is the blind origin, and the latent flow contracts toward it until some step reaches zero. *Proof.* As E8.2-E8.3; on the region the gradients are n_x (H_+ u_+ + H_- u_-) and n_y (H_+ u_+ - H_- u_-) with

H_+- positive definite, vanishing iff z_x = z_y = 0. [] For +1/-1 every channel's residual is the clock k_c >= 0, so the |.| latent gradient that removes the clock is a contraction toward the wall, while the signed one projects onto u = 0 and leaves the map alone (`prop_checks.py` (2): signed ends at |u| / |z| < 1e-3 with the map part unchanged; |.| ends at the blind origin).

E8.5 (rank; the clock shares every channel at rank D). One head, latent dimension r, move latents in R^r; for a D-dimensional map write the map frame U = [d_1 ... d_D] in R^{r x D} (half-differences of inverse pairs), rank U = D, and the common clock c (the latent step shared by every move, plus the mean observation step). Channel i reads a phase per unit displacement k_i = U^T f_i in R^D (f_i its readout row in diag(omega) W_out) and a drift tau_i = per move. (a) r = D, U invertible: c = U beta for a unique phantom displacement beta (cells per move), and tau_i = in every channel. A channel is drift-free iff its wavevector is orthogonal to beta, i.e. iff it does not resolve the drift direction. So while c != 0 no channel is both drift-free and a full D-dimensional map, and the only way to a clean code is c = 0 (the collective route). (b) r >= D + 1: split c = c_par + c_perp with c_perp orthogonal to range(U). If c_perp != 0, the readout change delta = -tau_i c_perp / |c_perp|^2 cancels channel i's drift with U^T delta = 0 exactly: the map is untouched to all orders (*parking*). Independent clock directions can be parked only up to r - D of them: on the torus a single common clock needs r >= 3, two independent pair clocks r >= 4. (c) At r = D the minimum-norm cancelling readout change sets k_i <- k_i - (k_i . bhat) bhat: it removes the channel's resolution along the drift direction (*hiding*). (d) At D = 1 hiding sets k_i = 0 (the channel dies), and the drift a channel sees in cells, tau_i / k_i = beta, is the same for every channel and every readout: no readout change alters the drift-to-map ratio; only shrinking c or growing U does.

Proof. (a) c in R^D = range(U) when U is invertible; = . (b) U^T c_perp = 0 by definition; for q independent clock directions solve = -tau_j, a linear system with independent coefficient vectors. (c) = ; Cauchy-Schwarz; the minimiser has U^T delta parallel to beta. (d) D = 1: k_i = U f_i, tau_i = c f_i, ratio c / U. [] Per head, because each head's channels read only that head's r-dimensional image: a block-diagonal rank-4 latent gives each head rank 2 and no c_perp. E8.5 and E8.2 use different clocks (the common step c, and the per-pair sums u); both vanish on a map code.

E8.6 (route inventory, stated once). For an inverse-pair map task, a class reaches the zero-clock set from a generic small start through the collective route iff its clock is a linear function of one latent vector whose zero set contains non-blind points, and through the per-channel route iff each channel has a clock-free, full-map readout reachable without crossing a fold. Then: signed, r >= D + 1: both (E8.2, E8.5(b)); signed, r = D: collective only (E8.5(a)); signed with frozen move latents: per-channel only (each channel can still cross zero through W_out); |.|: neither from a small start (E8.3: the latent route ends at a clock and no latent choice wraps more than 2r channels; E8.1 + E4.1: the per-channel zero-clock points lie at |k| = 2 pi n, across |k| = pi). A hybrid with readout |w_c| and signed latent, a_c(x) = omega_c |w_c| . z_x, has the signed class's collective route (E8.2 with W replaced by |W|) and its near-side cancelling point k = 0.

E9. Hiding at rank D (Proposition 5; Conjecture 1)

Assumptions. (A1) Small drift: for the gaps that carry gradient, g |tau_i| << 1, so the drift loss is L_d = (1/2) sum_i a_i tau_i^2 with a_i > 0. (A2) The map loss depends on the

frame only through the spectrum of the head's positional precision P = sum_i A_i k_i k_i^T, L_m = sum_j f(lambda_j(P)) with f decreasing and convex, plus a norm cost on readout rows. (A3) U near-isometric on the readout's span, so rotating k_i preserves |f_i|. Write r_beta = (bhat^T P bhat) / (tr P / D), the head's resolution along its own drift direction relative to its mean.

E9.1 (two routes). A change of parameters lowers tau_i only by changing c (shared by every channel of the head; the collective route) or by changing f_i with = -tau_i (the local route); E8.5 says what the local route costs.

E9.2 (Proposition 5; proved under A1-A3). (i) D >= 2, P isotropic: rotating any channel's k_i by a small angle eps from bhat toward a direction v orthogonal to bhat changes L_m by 0 at first order (a rotation leaves tr P unchanged) and changes L_d by -eps a_i |k_i|^2 |beta|^2 sin(2 phi_i) / 2 < 0 for every channel not already aligned with or orthogonal to bhat. So gradient flow on the readout moves resolution off the drift axis at once, for any beta != 0. (ii) D = 1: the only readout move that lowers L_d is shrinking |k_i|, whose first-order map cost is -f'(lambda) A_i > 0; it does not change beta (E8.5(d)). (iii) r >= D + 1 with c_perp != 0: parking lowers L_d at rate a_i |c_perp|^2 per channel with zero change in L_m at all orders. *Derivation.* k_i = |k_i| (cos phi bhat + sin phi v), tau_i = |k_i| |beta| cos phi, and d/deps (a_i / 2) tau_i^2 = -a_i |k_i|^2 |beta|^2 cos phi sin phi. []

E9.3 (Conjecture 1). Under A1 the gradient of L_d with respect to beta is P_a beta (P_a = sum_i a_i k_i k_i^T), so the collective route removes the drift along bhat at a rate proportional to bhat^T P_a bhat, the quantity hiding lowers. At rank D >= 2 this is a positive feedback (drift, then hiding, then slower removal, then persisting drift) that leaves the drift in place or produces a long plateau; the clean state (beta = 0, P isotropic) remains a stable fixed point, which is the projection that the rank-2 existence test installs and training holds. Two refinements: long gaps carry the loss but not the gradient (once g |tau_i| > 1 the cosine averages out), so drift removal is driven by short gaps while the failure shows at long gaps and wrap-only revisits; and the CLOCK and COLLAPSE end states of rank-D heads (partial and complete hiding) are points on one curve. *Status.* The first-order statement is proved; the feedback is a conjecture. A CPU pilot at T = 128 (n = 3) saw a failing rank-2 seed whose weak axis was present at initialisation, and at initialisation on the real task the drift is too large for A1 (App. G6).

E10. Exists, kept, not found (Remark 2; definitional)

Let Phi_B be the stochastic training map for budget B, S_eps = {theta : L(theta) <= eps} the solution set of a class and P_0 the initialisation distribution. Define the basin Bas = {theta : P(Phi_B(theta) in S_eps) >= 1/2}. *Exists:* S_eps is non-empty in the class (E2, constructed). *Kept:* the constructed points lie in Bas (ring 8/8, torus re-found 8/8, rank-2 projection 7/8). *Not found:* P_0(Bas) is small (SOLVED fractions 0-2/8 estimate it). Together they place the obstruction in the position of supp(P_0) relative to Bas, not in the emptiness or instability of S_eps. Two measurements of that position are in App. H7: the loss along straight paths from the start to a constructed solution, and the basin's extent along such a segment.

E11. Readers and codes (Proposition 6; proved)

E11.1 (one layer is a position-only reader). In a one-layer MapWM the pre-softmax score between the query at i and the key at j, head h, is s_h(i, j) = sum_c |q_c(x_i)| | k_c(x_j)| cos(theta_c(i) - theta_c(j) + psi_c(x_i, x_j)) / sqrt(d_head), where q_c, k_c are 2D blocks of W_q LN(e(x_i)), W_k LN(e(x_j)) and psi_c the angle between them. Both depend only on the token identities; position enters only through theta(i) - theta(j); the values depend on identity alone. So for scored revisit queries a one-layer model is a channel-weighted cosine kernel with token-type-dependent weights and offsets, a soft vote over the observations it lands on, and an MLP on the vote.

E11.2 (extra passes widen the reader, not the code). `MapFormerWM.forward` computes delta = action_to_lie(token_emb(x)) and cos, sin = path_integrator(delta) once and hands them to every layer; `MapFormerWM_Looped.forward` does the same and applies one block n_loops times. From the second pass on, q and k depend on the residual stream, so the weights |q_c||k_c| and offsets psi_c become context-dependent and values can carry retrieved content; theta is unchanged. A loop with the same parameters and initialisation is therefore the same code family with a larger reader function class. []

E11.3 (a score rule acts on the readout term). Any score of the form sum_c A_c cos(dphi_c + beta_c), with A_c >= 0, sees position only through dphi. Write the target coherence (the A-weighted cos(dphi + beta) on the target key) as C = C0 + (C - C0), with C0 the A-weighted cos(dphi) on the target, the code term. A score rule changes A and beta, so it changes C - C0 and can reweight channels within C0, but cannot change dphi. []

E12. Peak pinning (Proposition 7; proved given delta = 0)

RoPE / MapWM score: s_ij = sum_c Re(q_c conj(k_c) e^{i dphi_c}), so A_c = |q_c||k_c| and beta_c = arg q_c - arg k_c, both content-dependent. PoPE / MapPoPE (`model_pope.py`, cos(dphi - delta)): A_c = softplus(q_c) softplus(k_c), beta_c = -delta_c, content-free. If beta_c = 0 for all c, then for any content pair s_ij <= sum_c A_c, with equality iff dphi_c = 0 mod 2 pi on every channel with A_c > 0; as a function of displacement the kernel is a non-negative combination of cosines, positive-definite by Bochner, so content can reweight the frequencies but cannot move or invert the peak. *Corollary (index phase).* With PoPE's frequencies theta_c = 10000^(-c/64) <= 1 rad per token, each cos(theta_c D) decreases on D in [0, pi], so with beta = 0 every content pair's kernel strictly decreases over offsets 0-3: a one-layer pinned-peak index head cannot prefer the observation 3 tokens back to the one 1 token back, while the one-layer torus solution needs peaks at the retrace offsets 4j - 1. *Assumption check:* delta is in the clamp's dead zone on 98-100% of entries in every PoPE run (App. L).

E13. The used-channel ratio (Remark 3; light theory)

With token frequencies f_w, split every step into a common part and a content part: Delta_c(w) = mu_c + d_c(w), mu_c = sum_w f_w Delta_c(w), so sum_w f_w d_c(w) = 0. Over an interval of L tokens, Phi_c(t) - Phi_c(t') = mu_c L + (sum of d_c over the interval). The first term is a clock (Case I if carried by every token, Case II if only by selected tokens); the second can be a map (Case III) only if it cancels over closed loops. Define rho_c = |mu_c| L / range(content_c, L) at the training length L. E7.2 constrains mu_c only

in channels the readout uses. Prediction: in used channels rho_c << 1 when targets are keyed to a latent state reached at variable lags, and rho_c >> 1 when they are keyed to a token or event count. The pooled growth exponent of the summed step weights channels by their raw range, so a large tick in unused channels dominates it; the used-channel median of rho does not have this problem.

E14. Status

#	statement	status	check
E1	rotary position factor = Howard's propagator with s = i omega	proved (identity: Mamba-3 Prop. 3)	`a0_propagator.py`
E2 (a), (f), (g)	exact non-negative (	.	, softplus, exp) representation mod 2 pi with <= r stepping tokens
E3	relative-sign gauge; excluded codes are opposite-sign rows	proved	`t1_gauge.py`
E4	cut circle; sticky wall along s	proved	`p3_p5_geometry.py`
E5	CARoPE: every channel a clock or blind	proved	`p6_carope.py`
E6.1	prefix sum in the phase; one layer at any gap	proved	same-cell attention (App. F2)
E6.2	two-layer global count (normaliser, precision, width)	cited (Liu et al. Lemma 6; Yao et al.)	probes (App. F3)
E7	phase decomposition; position-factor bound; tick rule	proved	bound consistent with the strata, not a test of them; text-world drift (App. J)
E8.1-E8.4		.	clock floor; signed collective route; no
E8.5	rank D: no clock-free full-map channel; rank > D: parking	proved	end states (App. G)
E8.6	route inventory	proved (inventory)	Table 4; SIGNROUTE
E9.2	hiding free to first order at D >= 2	proved under A1-A3	drift on the least-resolved axis (App. G)
E9.3	hiding starves the collective route	conjecture	not tested
E10	basins	definitional	App. H7
E11	readers and codes	proved (architecture, algebra)	one-pass reader (App. M)
E12	peak pinning	proved given delta = 0 (measured)	DELTA_FREE (App. L)
E13	used-channel ratio	light theory	App. J4

Statements of the theory appendix v3 not used here: P1(c)-(e), P2 (rescaling sensitivity; its drift account is withdrawn), P5 (the pi-mirror) and P7 (the CoPE lattice); they are in `docs/paper/APPENDIX_theory_v3.md` with their check scripts.

F. Exchange-rate details

Dyck-2 (`DYCK_MDEPTH_RESULTS.md`). Task and metric as MapFormer v4 App. B.4, trained and scored at length L = 32 and nesting depth D = 12; primary metric A2f, Hewitt's distance-averaged closing accuracy over prefixes where a closer can be emitted (CE-optimal predictor 1.000, chance 0.500, best stack-free floor 0.594). 2 heads, d_model 128, 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2; index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); n = 8 paired seeds. Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2.

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	1	0.648	0.640	0.996	0.998	+0.353
1x budget	2	0.889	0.850	1.000	1.000	+0.130
1x budget	3	0.964	0.946	1.000	1.000	+0.045
1x budget	4	0.982	0.971	1.000	1.000	+0.024
3x budget	4	0.998	0.997	1.000	1.000	+0.002
mixture of depths 4-12	4	0.898	0.881	0.999	0.999	+0.110

Every effect is positive on 8/8 seeds (sign-flip p 0.0078). Index base 32 minus base 10000 at four layers, 3x budget: +0.001 (RoPE), -0.001 (PoPE), within the MDE. Three times the budget raises the index arms by +0.021 at four layers (8/8).

Ring (C3). The index-one-layer accuracies are not monotone in p_plus (0.710, 0.677, 0.824, 1.000 on fresh seeds; every pair differs, p 0.0022-0.0108), so the exchange rate, not the one-layer accuracy, is the quantity that tracks whether steps are undone.

Torus (`FACTORIAL_RESULTS.md`, Fig. F1). Strata (short gap / long gap / wrap-only; mean and run range, n = 4): path one layer 1.000 / 1.000 / 1.000; index one layer 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); four layers 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521). Floors on this stream (analysis): constant 0.508 / 0.515 / 0.517, retrace 0.912 on short gaps. Memorisation check: four index layers 0.897 on their own training map against 0.860 held out; looped 0.862 against 0.864; path 1.000 on both.

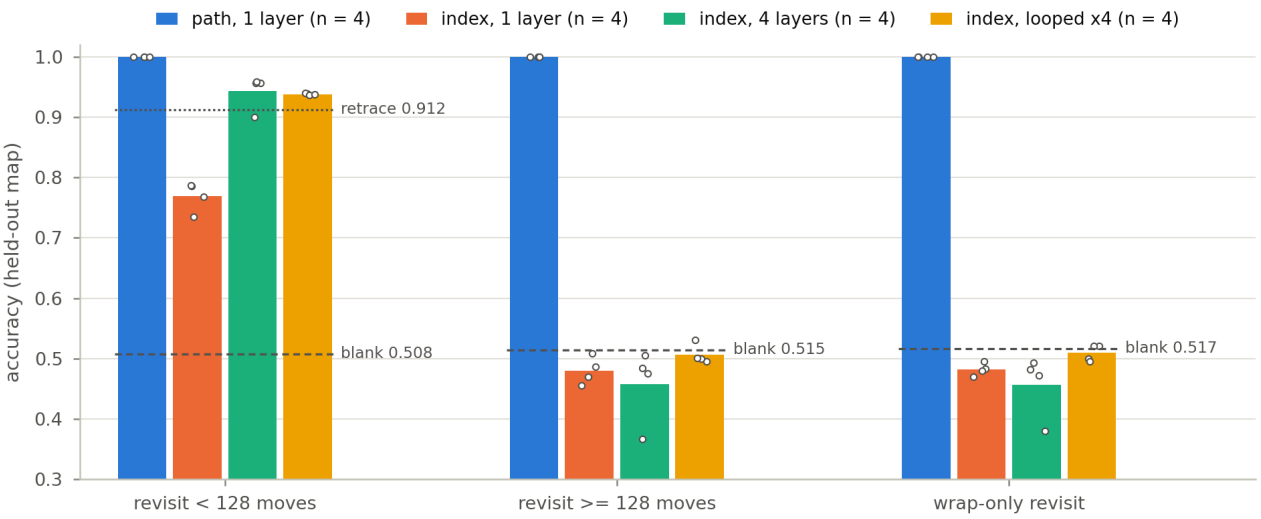


Figure F1 (`fig_factorial_strata`). Revisit accuracy by stratum on the torus at T = 1024 (trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Dashed: always-blank within each stratum, 0.508 / 0.515 / 0.517; dotted: retrace within the short-gap stratum, 0.912 (elsewhere it equals the constant); floors on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

F1. Which revisits depth buys on the torus (analysis; `scripts/depth/torus_depth.py`, `summarize_torus.py` -> `out/torus_depth_out.txt`, `summarize_torus.py --segments` -> `out/torus_segments_out.txt`)

The FACTORIAL evaluation stream was regenerated (env seed 10000, np seed 1234 + s, 100 walks); overall accuracy per run equals the stored strata files' "all" to 4 decimals for all 16 runs, and the re-implemented forward equals the model's logits (max |diff| 0.0). Per revisit: s = the last visit to the cell; *retrace* = the current run reverses the previous one and has not overshot it; *runs* = maximal single-action runs among moves s+1..t. Means of per-seed accuracies (n = 4; range in brackets).

revisit class	share	index 1L	index 4L	loop x4	path 1L
short gap, retrace	0.680	0.833 (0.79-0.86)	0.996 (0.99-1.00)	1.000	1.000
short gap, not retrace	0.148	0.475	0.699 (0.51-0.77)	0.655	1.000
... 3-4 runs	0.031	0.477	0.926 (0.74-0.99)	0.977	1.000
... 5-8 runs	0.056	0.476	0.735 (0.52-0.81)	0.645	1.000
... 9-16 runs	0.052	0.474	0.561 (0.38-0.64)	0.499	1.000
... 17+ runs	0.009	0.472	0.493	0.502	1.000
long gap (>= 128)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

Gap in moves (plain revisits): index 4L / loop 1.000 / 1.000 at 2-7, 0.978 / 0.986 at 8-31, 0.614 / 0.551 at 32-127, 0.458 / 0.506 at 128-511; one layer 0.935 at 2-7, 0.681 at 8-31, floor beyond. Runs at a fixed gap of 32-127 moves (plain revisits, retraces included; mean of per-seed accuracies, n pooled over the 4 seeds; the 2-run cells are all retraces of merged runs; pooled over seeds instead, the small cells read 4L 0.896 at 2 runs and 0.785 at 3-4):

runs spanned	n	index 1L	index 4L	loop x4
2	240	0.412	0.874	0.996
3-4	503	0.483	0.832	0.911
5-8	4565	0.464	0.688	0.575
9-16	6569	0.474	0.560	0.499
17+	1202	0.472	0.493	0.502

At gaps of 8-31 moves: 4L 0.993 / 0.940 / 0.818, loop 1.000 / 0.987 / 0.768 at 2 / 3-4 / 5-8 runs. Long-gap revisits span 17 or more runs in 98.5% of cases (median 37, minimum 11), wrap-only revisits in 99% (median 65). The deep models score below the constant floor there with high loss (2.1-4.8 nats): they predict a specific wrong object rather than

abstaining to blank. *Decomposition* (an identity, since short = retrace + not retrace): the short-gap retrace floor is (0.680 x 1 + 0.148 x ~0.5) / 0.828 = 0.911 (measured 0.912); four index layers give (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943. The class contrasts at n = 4 (e.g. 4L against the loop at 3-4 runs; 4L range 0.74-0.99) carry no test.

F2. Where index attention looks (analysis; same script)

Attention at the revisit query (token a_t), final pass, per head; *same* = mass on earlier observation tokens of the same cell (uniform chance 0.001-0.003); *last* = mass on the last visit's observation and arrival tokens.

model, final pass	retrace: same / last	short, not retrace: same	long gap: same	wrap-only: same
index 1L	0.05 / 0.10 (h0), 0.06 / 0.12 (h1)	0.00	0.00	0.00
index 4L, layer 4	0.55 / 0.56, 0.58 / 0.58	0.22 / 0.22	0.01 / 0.01	0.00 / 0.00
loop x4, pass 4	0.50 / 0.52, 0.49 / 0.50	0.20 / 0.20	0.00 / 0.00	0.00 / 0.00
path 1L	0.83 / 0.70, 0.76 / 0.65	0.82 / 0.76	0.81 / 0.74	0.66 / 0.58

In the deep index models retrieval happens in the last pass only, and on retraces all of it is on the last visit: they find the mirror point, not the cell.

F3. No global count on the torus (analysis; `scripts/depth/torus_probe.py` -> `out/torus_probe_out.txt`)

Ridge probes from the residual stream at move tokens (train walks 0-29, test 30-39, t >= 16; out-of-sample R^2, mean of 4 seeds, x and y averaged; best pass):

target	index 1L	index 4L	loop x4	ring index 3L, p_plus 0.5 (t >= 8; see F5)
unwrapped displacement P_t	-0.06	0.10	-0.01	0.60
normalised count P_t / (t+1)	-0.03	0.21	0.06	0.69
1 / (t+1)	-0.01	0.78	0.74	0.89
displacement over the last 4 moves	0.90	0.96	0.95	0.97
... last 16	0.47	0.93	0.86	0.86
... last 64	0.09	0.77	0.56	0.70
... last 256	-0.01	0.37	0.14	--

The deep torus models know elapsed time (1/t) but carry no normalised count; they carry a recent-displacement code whose error grows with the window (0.5 cells at 4 moves, 1.8 at 16, 7.1 at 64, 25 at 256 for 4L; the d_w targets at t < w are the displacement since the start, so the rmse at w = 256 is approximate). Linear probes bound only what is linearly readable; the attention readout of F2 does not rely on them. The ring column's count-like R^2 is a window code (F5).

F4. Run length moved at evaluation (analysis, distribution-shifted; `scripts/depth/torus_runlen.py`)

The stored FACTORIAL models re-evaluated on walks whose run length is fixed at 2, 5 or 10 moves. Two-run returns are gap-invariant for deep index models (retraces at gaps of 16-63 moves with runs of 10: 0.978 / 0.995 for 4L / loop). Down a column of fixed gap (16-63 moves) 4L falls from 0.98 (2 runs) to 0.87-0.93 (3-4), 0.72-0.80 (5-8), 0.64-0.70 (9-16) and 0.58 (17+); within a run bin across gaps it moves by 0.02-0.2. Runs set the level and the gap trims it. Path 1.000 in every cell of every condition.

F5. The ring: rules, runs and a count (analysis; `scripts/depth/ring_depth.py`, `summarize_ring.py`)

CANCEL evaluation stream (overall accuracy equals each run's stored value for all 128 runs); fresh seeds 2-7. Classes: *retrace*; *lap* = the displacement over the last 32 moves is 0 mod 32 (a fixed one-lap offset lands on the cell); *neither*, split by runs. p_plus 0.5, mean (range):

class	share	index 1L	index 2L	index 3L	path 1L
retrace	0.716	0.797	0.987	1.000	1.000
lap, not retrace	0.035	0.497	0.977	1.000	1.000
neither, 3-4 runs (wrap-only included)	0.127	0.487	0.891 (0.72-0.94)	0.997	1.000
neither, 5-8	0.097	0.489	0.802 (0.55-0.87)	0.993	1.000
neither, 9-16	0.025	0.485	0.678 (0.48-0.75)	0.972	1.000
wrap-only	0.118	0.503	0.819	0.989	1.000

The "neither" rows include wrap-only revisits (29% / 47% / 56% of the three bins); on plain revisits only, 2L scores 0.910 / 0.828 / 0.742 and 3L 0.999 / 0.996 / 0.988 (`ring_plain_bins.py`; the conclusions, 2L decays and 3L is flat, are the same). p_plus 0.75 / 0.9 (index 1L): retrace 0.692 / 0.677, lap 0.987 / 0.994, neither 0.487 / 0.592, against best-constant floors of 0.493 / 0.491 / 0.502 for the "neither" class at p_plus 0.5 / 0.75 / 0.9 (`ring_class_floors.py`): at 0.9 one layer sits an unexplained +0.09 above it. At p_plus 1.0 every revisit is a lap revisit, 1.000 at every depth. *No global count on the ring either* (`ring_probe_window.py` -> `out/ring_probe_window_out.txt`). Probes read at t >= 8 on 128-move walks (P_t R^2 0.60, P_t / t 0.69, 1/t 0.89 for three layers; one layer 0.09 / 0.18 / 0.14; two 0.40 / 0.60 / 0.67) cannot separate a prefix count from a recent-window code, since half of P_t / t's variance sits at t <= 32 and for t < 64 the 64-move window equals P_t. Read late, three layers give P_t R^2 0.59 at t >= 64 (two layers 0.32), and at t >= 96 the displacement before the last 64 moves has R^2 0.01 (before the last 32: 0.31), while the windows are well coded (last 64 moves 0.72); the true 64-move displacement alone predicts P_t there with R^2 0.51, against the model's 0.53. Three ring layers carry a displacement code reaching 32-64 moves back, which on 128-move walks covers most revisits; their last-layer attention reaches the cell on wrap-only revisits (same-cell mass 0.36 / 0.38). *One-layer accuracy across p_plus (descriptive):* a p-invariant mixture does not reproduce it: per-class accuracies pooled over p predict 0.678 / 0.683 / 0.749 against the observed 0.710 / 0.677 / 0.824 at p_plus 0.5 / 0.75 / 0.9, because the class accuracies themselves move with p (retrace 0.797 -> 0.692 from 0.5 to 0.75, as large as the net dip of 0.033). No mechanism is claimed for the dip at 0.75.

F6. Dyck-2: blocks to skip (analysis; `scripts/depth/dyck_distance.py`, `dyck_crosstab.py`)

Closing accuracy by the number of complete balanced blocks between the top opener and the prefix (mean of 8 seeds):

blocks to skip (n prefixes)	RoPE 1L	2L	3L	4L	MapWM 1L
0 (3682)	0.998	0.999	0.999	0.999	0.999
1 (5839)	0.640	0.925	0.984	0.991	0.997
2 (572)	0.641	0.778	0.885	0.934	0.993
3-4 (161)	0.613	0.760	0.860	0.925	0.993

Dyck has a BOS token, so the count route's normaliser is available, and both blocks and distance matter for the index models; the path model is flat in both (0.987-0.998). 93% of scored prefixes need at most one block, which is why three index layers already reach 0.964 in aggregate.

G. Rank details

G1. Arms, curves and dimension

Arms (A1). Shared r2 (`Vanilla`), per-head r2 (`Vanilla_r2ph`), block-diagonal r4 (`Vanilla_r4mibd`: the shared rank-4 latent with W_out's off-diagonal blocks held at zero), per-head r3 (`Vanilla_r3ph`), shared r4 (`Vanilla_r4mi`), per-head r4 (`Vanilla_r4ph`), all built from the shared rank-2 model's initial weights at each seed. Initial angle-increment scale (std): 0.208 for block-diagonal r4 against 0.33-0.34 for the other arms, which fail (rank 2) or solve (rank 4) from the same scale. Per-seed results: C4.

Training curves (Fig. G1; `fig_loss_curves_rank_out.txt`). Shared r2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final loss 0.35) against shared r4 8/8 (2.8e-3); per-head r2 2/8 (median 0.22), r3 6/8 (3.5e-3), r4 8/8 (3.1e-3). The rank-3 and rank-4 medians cross the threshold at epochs 478-532; the rank-2 medians never do.

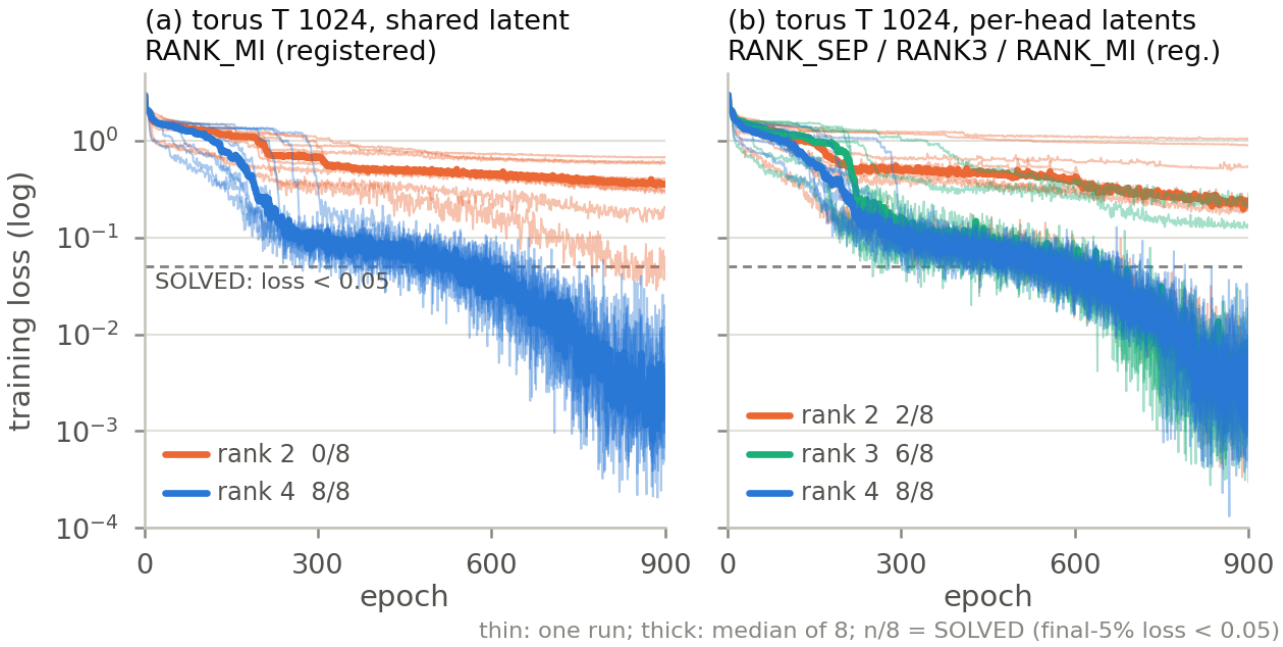


Figure G1 (`fig_loss_curves_rank`). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8. (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8.

Restart curves (`RANK_PROJ_ANALYSIS.txt`). Projected rank 2: peak training loss 0.37-0.46 after the restart, back below 0.05 at epochs 553-654 on 7/8 seeds. Rank-4 control: peak 0.26-0.30, back at 497-669 on 7/8. Rank 2 continued: peak 0.55-0.82, back on 1/8.

Rank against dimension (`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`). Per-head rank, every arm from the same base, trained and tested at T = 1024, 900 epochs:

torus	D	per-head rank	wrap-only share of revisits	SOLVED	held-out accuracy
32 per side	2	2	0.37	1/8	0.769
32 per side	2	3	0.37	8/8	0.999
10 per side	3	3	0.71	1/8	0.800
10 per side	3	4	0.71	4/8	0.847
18 per side	3	4	0.35	8/8	1.000

2D rank 3 vs rank 2: 8/8 vs 1/8 (Fisher p 0.0014), +0.230 (permutation p 0.0014). 3D, 18 vs 10 per side at rank 4: +0.153 (p 0.0085). On a 100-cell 2D torus (10 per side) rank 3 memorises its training map (0.986 on it, 0.273 on an unseen map), so that cell does not measure map learning. 1D ring (32 cells, T = 128, 300 epochs): rank 1 7/8 (0.993), rank 2 8/8 (1.000); with the component common to the move steps removed by construction, rank 1 solves 8/8 in a median of 24 epochs (58 without).

Escape test (`RANK_TESTS_RESULTS.md`). Lifting the eight stalled shared-rank-2 runs to rank 4 (function-preserving) and training 600 epochs solves 3/8, against 1/8 for rank 2 continued

and 2/8 for a fresh rank-4 step map on the same networks; at this budget the contrasts are not resolved.

G2. PoPE's score does not rescue rank 2 (`SCORE_RANK_RESULTS.md`, registered NO RESCUE)

Torus at T = 1024, 900 epochs, 32 angles per head in every arm; rank-2 arms on seeds 10-21, rank-4 arms 10-17:

arm	accuracy	SOLVED
MapWM r2	0.848 +/- 0.178	3/12
PoPE score r2	0.946 +/- 0.040	3/12
MapWM r4	0.9998	8/8
PoPE score r4	0.9995	8/8

Primary: 3/12 vs 3/12 (Fisher p 1.00); the rank-2 deficit replicates on these fresh seeds under both scores (8/8 vs 3/12, p 0.0014 each). The stratum readouts are secondaries (short gaps +0.115, p 0.036; gap >= 128 +0.043; wrap-only -0.015).

G3. The clock at the end of training, per head (analysis; `scripts/sign/clock_table.py` -> `clock_table_out.txt`; latent column `scripts/sign/rank_clock.py` -> `rank_clock_out.txt`)

Clock / map ratio of Sec. 4 (E7.1') per run (usage pooled over both heads) and per head (usage normalised within the head); full models (observations step). The latent clock (|z_x + z_y| over the map frame, raw latent coordinates, mean over unsolved runs' heads) is a different quantity and not invariant to the GL(r) gauge of the latent (in the W_out image the unsolved values become 1.4-2.6).

arm	SOLVED	solved runs: run mean (heads median, range)	unsolved runs: run mean, median (heads median, range)	latent clock, unsolved
shared r2	0/8	--	0.595, 0.579 (0.619, 0.07-1.26)	1.72
per-head r2	2/8	0.236 (0.045, 0.03-0.54)	0.694, 0.792 (0.756, 0.02-1.98)	3.19
block-diagonal r4 (rank 2 per head)	2/8	0.335 (0.067, 0.04-0.68)	0.541, 0.575 (0.084, 0.01-1.41)	1.63
per-head r3	6/8	0.043 (0.038, 0.02-0.08)	0.936 (0.678, 0.12-1.25)	3.68
shared r4	8/8	0.049 (0.036, 0.01-0.10)	--	--
per-head r4	8/8	0.051 (0.030, 0.02-0.14)	--	--

The solved rank-2 run means hide a split: in each of the four solved rank-2 runs at least one head is clean (per-head r2 s1 h1 0.03, s2 0.05 / 0.05; block-diagonal s2 0.04 / 0.06, s6 h0 0.07), and two runs keep a drifting second head (per-head s1 h0 0.54, block-diagonal s6 h1 0.68). Not every failure is clock-coupled: 5 of the 20 unsolved rank-2 runs have both heads below 0.15 (shared s1; per-head s5, s6; block-diagonal s1, s7), and 4 of them (all but shared s1) are in the collapsed class of `rank/sweep_final_out.txt`: visible drift <=

0.01 per move with the two map axes near-collinear (frame conditioning 0.00-0.03), a map that has lost a dimension; shared s1 ends just above the SOLVED threshold (final loss 0.051). Heads below 0.15 among unsolved runs: shared 4/16, per-head 4/12, block-diagonal 7/12. The rank-3 unsolved runs are clock-coupled. The observation clock (mean observation step over the map; `rank_clock.py`) is 0.015-0.018 in solved rank-3/4 heads and 0.18-0.52 in unsolved heads. At r = 2 = D, rank_clock's "in-span" share of the clock is 1 by construction before wrapping, so it is not reported as a measurement.

G4. The leftover latent clock and hiding (analysis; `scripts/rank/`; a latent quantity, not the ratio of G3)

Here the clock is c = a + o, the mean move latent a plus the mean observation latent o (the drift a walk accumulates per move and observation), measured in the native latent of each head's read subspace against the map frame U (lam = |c| / |U|); "off-plane share" = |c_perp| / |c|; "readgain" = how strongly the readout sees the leftover clock relative to the map (1 = read like the map, 0 = invisible). This latent metric and G3's channel ratio agree that what is still read in solved heads is small; they are not numerically comparable. The native latent metric is not invariant to the GL(r) gauge of the latent; the independent audit found the balanced gauge gives the same picture (solved-set medians 0.94-1.00).

What the leftover clock is made of (`clock_decomp.py` -> `clock_decomp_out.txt`; medians over solved heads): a and o are each 0.017-0.037 of a map step and anti-parallel (cosine -0.79 to -0.999), so they cancel; at T = 1024 they lie mostly in the map plane (off-plane share 0.05-0.35 for a, 0.02-0.08 for o). Only their small residual c is off-plane. SEARCH's per-pair clocks z(-e_j) + z(+e_j), which leave the observation step out, are 0.034-0.057 of a map step and in-plane (off-plane 0.14-0.21 on the 2D sets, below the random-direction median); with the two observation steps of one out-and-back added, 0.73-0.99 off-plane. The per-pair clock therefore overstates the drift a walk accumulates; the two definitions agree on what is read.

set (solved heads)	|a| / |U|	|o| / |U|	cos(a, o)	off-plane share of a / o / c
torus G64 per-head r3	0.021	0.020	-0.988	0.15 / 0.07 / 0.98
torus G64 shared r4 (RANK_MI)	0.017	0.017	-0.980	0.19 / 0.08 / 1.00
3D G18 r4	0.021	0.022	-0.999	0.05 / 0.06 / 0.81
torus T = 128 r4	0.034	0.029	-0.897	0.48 / 0.40 / 0.99
ring rank 2	0.037	0.032	-0.793	0.47 / 0.17 / 0.99

Where the residual lies in solved heads of rank above D (`sweep_final.py`, `summarise_final.py`; consistent with parking, but removal through the latent, which takes the read components first, leaves the same end state, so this does not discriminate the two):

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
torus G64 per-head r3	1	0.004	0.978 (0.50)	0.15
torus G64 shared r4 (RANK_MI)	2	0.004	0.996 (0.71)	0.11
torus G64 per-head r4	2	0.003	0.988 (0.71)	0.16
	2	0.003	0.998 (0.71)	0.09

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
torus G64 shared r4 (SCORE_RANK)				
2D G32 r3	1	0.003	0.987 (0.50)	0.09
3D G10 r4	1	0.008	0.964 (0.40)	0.23
3D G18 r4	1	0.001	0.810 (0.40)	0.57
ring rank 2 (1D), 300 epochs	1	0.025	0.989 (0.71)	0.21
torus T = 128 r4, 300 epochs	2	0.021	0.988 (0.71)	0.27
escape, fresh r4 step map, 600 epochs	2	0.003-0.004	0.98 (0.71)	0.18

In the clean head of solved rank-D runs (ring rank 1, torus rank 2) the clock is removed (lam <= 0.0005 on the T = 1024 2D sets, <= 0.0013 on the ring), since it cannot lie off the plane; the other head of such runs often drifts (lam 0.15-1.75). The leftover is about 6x larger after 300 epochs than after 600-900, of the size decoupled weight decay alone gives an unread direction over a cosine schedule (consistent, not a test). The escape arms (below) show the latent clock shrinking, not held: in the fresh rank-4 step maps its absolute size falls 14-85x (relative to the map step 270-1300x), the off-plane part 15-72x in 7/8 runs (1.6x in s1, which started at 0.005), and in the three solved lifted runs 23-490x (`clock_decomp_out.txt`). Unsolved heads of rank above D read their clock in all three sets and hold it in the map plane in two (torus per-head r3: off-plane 0.27 against a random-direction median of 0.50, readgain 0.61; escape lifted runs: 0.08 against 0.71, readgain 0.76; 3D G10 r4: off-plane 0.53 against 0.40, readgain 0.78).

Drifting rank-D heads drift along their least-resolved axis (`c1_runlevel.py`): over 116 drifting heads in nine sets the median r_beta (resolution along the drift direction relative to the head's mean) is 0.087, and 85% are below 0.5 against 29% for a random direction given each head's own frame; at the run level 57 of 62 runs against 28.7 expected (P < 5e-5, 20,000 simulations); escape C continues rank_mi's own networks, so counting each network once gives 50 of 55 against 25.1 (`c1_runlevel_once.py`). Clean heads of rank above D sit at r_beta 0.79-1.10 (medians). This geometry is what E9 leads to and also what pruning of a visible drift after any collapse would leave; it fixes the geometry, not the order of events.

The escape test, re-read (`escape_start.py`; RANK_TESTS escape arms, 600 epochs on the same eight trained networks):

arm	runs with a drift-free head at the end	heads drift-free	off-plane share at end
fresh rank-4 step map (start drift 0.28-5.86 cells per move)	8/8	16/16	0.91-1.00
stalled rank 2 lifted to rank 4	3/8	6/16	1.00 in the 3 solved
rank 2 continued	1/8	2/16	--

Within the lifted arm the clock's off-plane share at the start (the lift adds two latent rows at default scale beside large trained rows) is 0.09 / 0.14 / 0.11 in the three runs that solve

and 0.01-0.06 in the five that do not (exact one-sided rank-sum p 0.018; post hoc). Two caveats make this suggestive only. The statistic was chosen after the fact: with the off-plane clock relative to the map step instead (|c_perp| / |U|), the solved runs rank 1st, 2nd and 8th (p 0.29). And s1's source also solves under plain continuation (arm C s1 SOLVED) and starts nearly clock-free (|c| / |U| 0.05), so its large share is that of a near-zero clock; without s1, p 0.048 (`escape_start_out.txt`). A seeded escape test (RANK_THEORY's G2) would test it. The fresh step maps remove the drift everywhere, and the residual geometry does not track SOLVED within that arm (the 12 unsolved heads show the same off-plane share, 0.985, and readgain, 0.19, as the 4 solved ones; audit); learning the rest of the map within 600 epochs (2/8 SOLVED) is a separate matter.

G5. The 1D exemption (registered + analysis)

On the ring (`RANK_TESTS_RESULTS.md`), rank 1 solves 7/8 (0.993) and rank 2 8/8 (1.000); with the drift forbidden by construction (opposite move latents tied, observation steps zero) rank 1 solves 8/8 in a median of 24 epochs (58 without). Every channel of a rank-1 head sees the same phantom drift in cells (E8.5(d)); over the 8 rank-1 runs the 11 clean heads (visible drift <= 0.01 per move) end at lam <= 0.0013, and the 5 drifting heads (visible drift 0.04-0.50 per move; two of them in the unsolved run s22) have the largest map steps (|U| 2.8-5.0 against 1.6-2.6), the only other global way to shrink the drift in cells.

G6. Dynamics (analysis, CPU pilot; `scripts/rank/train_snap.py`, `analyze_snap.py`, `pilot_t128/`)

Paper torus T = 128 (grid 64), batch 32, 40 epochs cosine, lr 1e-3, seeds 3000-3002 (outside every batch); arms shared r2, shared r4 (from r2's base) and R2Clock (shared r2 plus a fixed extra latent coordinate that every token steps by 1, giving each channel a learned clock rate; identical to r2 at initialisation). Snapshots every 49 steps.

arm	seed	final loss	accuracy T = 128 / 512	latent clock
r2	3000	0.005	1.000 / 0.937	removed in lockstep with the visible drift (lam 0.63 -> 0.004)
r2	3001	0.230	0.922 / 0.836	kept, on the unresolved axis (r_beta 0.00); frame near-collinear from step 0
r2	3002	0.003	1.000 / 0.981	removed in lockstep
r4	3000	0.005	1.000 / 0.954	off-plane 0.62 -> 0.96, then removed (lam falls 66x)
r4	3001	0.028	0.990 / 0.959	off-plane 0.55 -> 0.98, then removed (180x)
r4	3002	0.005	1.000 / 0.946	off-plane 0.81 -> 1.00, then removed (5.5x)
R2Clock	3000-3002	0.003-0.006	1.000 / 0.941-0.959	the slot holds a fixed unit clock by construction (off-plane ~1 from step 0)

At rank 4 the visible drift fell tenfold one or two snapshots before the latent clock did in 4/6 heads, and the latent clock was then removed too (5-180x within 40 epochs); in the rank-2 control the same order appears in 2/6 heads (n = 3). R2Clock is not counted: its slot clock cannot shrink tenfold by construction. At the same initial weights as the failing rank-2 seed 3001, R2Clock solved (loss 0.006 against 0.230) and repaired its frame (one matched pair). The failing seed's weak axis was present at initialisation; the readout

gradient at initialisation on the real T = 1024 task gives no consistent hiding direction (drift 0.4-2.7 cells per move, outside assumption A1; one stochastic sample of two batches, dropout on; `grad_hiding.py`). Initial frame conditioning does not separate solved from unsolved rank-D runs at T = 1024 (one-sided p 0.19; `init_frame.py`); initial drift is lower in the runs that solve (as a test only the independent SCORE_RANK seeds count: one-sided p 0.073; pooled over sets whose seeds share a base, 0.012, descriptive). MapFormer's 5D setting (T = 16, grid 5) solved at every rank under our recipe (rank 5, 6 and 5 + clock slot 3/3 each; index RoPE stalled), so it does not test rank here (`pilot_5d/`).

H. Sign details

H1. Learning rates (`LRSWEEP_RESULTS.md`; ring, observation-step-free, seeds 1500-1507)

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy
1e-3	8/8, 1.000	1/8, 0.917
3e-3	8/8, 1.000	2/8, 0.963
1e-2	4/8, 0.972	1/8, 0.872
3e-2	1/8, 0.559	0/8, 0.584

Registered B1 (each arm at its best rate, selection redone in the null over all 12,870 relabellings): |.| 3e-3 2/8 vs signed 1e-3 8/8, -0.037, decision p 0.0070 (SOLVED) / 0.0057 (accuracy). The |.| optimum is interior to the grid. Three |.| runs at 3e-3 were still descending at 300 epochs. At 3e-2 both arms sit near the 0.50 blank floor. Per seed:

```
seed | 1e-3:SO | 1e-3:AO | 3e-3:SO | 3e-3:AO | 1e-2:SO | 1e-2:AO | 3e-2:SO | 3e-2:AO
1500 | 1.0000 SOLV | 0.9362 STAL | 1.0000 SOLV | 0.9111 STAL | 0.9460 DESC | 0.9936 DESC |
0.4960 STAL | 0.4958 STAL
1501 | 1.0000 SOLV | 0.8759 STAL | 1.0000 SOLV | 0.9662 DESC | 0.9460 STAL | 0.4960 STAL |
0.4960 STAL | 0.4960 STAL
1502 | 1.0000 SOLV | 0.8995 STAL | 1.0000 SOLV | 0.9428 STAL | 0.9407 STAL | 0.9703 DESC |
0.4960 STAL | 0.4960 STAL
1503 | 1.0000 SOLV | 0.9116 STAL | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.6000 STAL |
0.4960 STAL | 0.4960 STAL
1504 | 1.0000 SOLV | 0.9415 STAL | 1.0000 SOLV | 0.9686 STAL | 0.9460 STAL | 1.0000 DESC |
0.4960 STAL | 0.8159 DESC
1505 | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.9993 SOLV | 1.0000 SOLV | 0.9950 DESC |
0.9998 SOLV | 0.4961 STAL
1506 | 1.0000 SOLV | 0.8736 STAL | 1.0000 SOLV | 0.9595 DESC | 1.0000 SOLV | 0.9996 SOLV |
0.4960 STAL | 0.4960 STAL
1507 | 1.0000 SOLV | 0.8971 STAL | 1.0000 SOLV | 0.9587 DESC | 1.0000 SOLV | 0.9184 DESC |
0.4959 DESC | 0.8799 DESC
```

H2. Recipe cells (`ROBUST_RESULTS.md`, seeds 1400-1407)

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
ring	lr 1e-3, weight decay 0.05	0.911, 1/8	1.000, 8/8	-0.089 [-0.125, -0.055]	0.0002	0.0014
ring		0.873, 0/8	1.000, 8/8		0.0002	0.0002

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
	lr 1e-3, weight decay 0			-0.127 [-0.200, -0.075]		
ring	lr 3e-3, weight decay 0.05	0.987, 4/8	1.000, 8/8	-0.013 [-0.025, -0.005]	0.026	0.077
ring	lr 3.3e-4, weight decay 0.05	0.841, 0/8	0.950, 6/8	-0.109 [-0.215, +0.010]	0.089	0.0070
torus	2 layers, d_model 256	0.967, 2/8	0.999, 8/8	-0.032 [-0.055, -0.015]	0.0005	0.0070

At lr 3e-3 the registered contrast was below its decision threshold in this batch (4/8 vs 8/8); on fresh seeds in the sweep the same rate gives 2/8 vs 8/8 (H1). Declared secondaries: cancelling share (unweighted) |.| 0.04-0.13 in every cell, signed 0.43-1.00; torus strata (short / long / wrap-only) signed 1.000 / 0.999 / 0.992, |.| 0.989 / 0.911 / 0.799; on the torus 6/8 |.| runs were still descending at 900 epochs (final losses 0.018-0.144).

H3. Starts (`MECHANISM_RESULTS.md`, ring p_plus 0.5, seeds 800-807; `TORUS_START_RESULTS.md`, torus, seeds 1200-1207)

Observation-step-free models differing only in the initial move angles (every other weight identical at a seed):

initial move angles	signed (ring)	|.| (ring)	signed (torus)	|.| (torus)
default (small, signs independent)	8/8	0/8	8/8	1/8
the |.| start: both moves rotating the same way	7/8	0/8	--	--
large, opposite-move sums below pi, U(0.9, 1.5)	--	2/8	--	0/8
large, sums above pi, U(1.7, 4.5)	8/8	8/8	2/8	0/8

Ring, registered: same start, sign matters (0/8 vs 7/8, Fisher p 0.0014; -0.092 [-0.135, -0.045]); a start past pi rescues |.| (8/8 vs 0/8, +0.109 [+0.085, +0.135], p 0.0002); a large start with sums below pi does not (2/8; vs past pi Fisher p 0.0070); from past pi the sign makes no difference (8/8 vs 8/8). The large start also moves the angles 13-22x faster per Adam step (the sub-pi control about 5x), so sum and size are only partly separated. Analysis: signed models from the small starts end with 98-99% of channels opposite-signed and cancel through zero; from past pi both signs cancel by wrapping (signed 88%, |.| 100% of cancelling channels); the rescued |.| models cancel in 23% of channels. Torus, registered: the past-pi start does not rescue |.| (0/8 vs 1/8, +0.005 [-0.060, +0.070]), and it slows the signed model (2/8; 5/8 runs per large-start arm still descending at 900 epochs); the default-start sign cost replicates (1/8 vs 8/8, Fisher p 0.0014). The start lever is a ring result.

H4. Moves +1/-1 vs +1/+2 (`GAUGE_RESULTS.md`, ring, seeds 1300-1307)

By Lemma 1 (ii) a |.| step represents, without wrapping or crossing zero, a map whose two moves step the same way. With moves +1/-1 (an inverse pair) the cost replicates: |.| 0.931, 1/8 vs signed 1.000, 8/8 (-0.069 [-0.095, -0.045], Fisher p 0.0014). Moves +1/+2 need integration (index RoPE 0.649, 0/8) and every revisit is a lap of at least 16 moves;

there |.| scores 0.883, 2/8 against signed 0.947, 5/8 (-0.064 [-0.200, +0.075], Fisher p 0.31), a registered comparison that is not resolved because the task is harder for both arms. The step code is learned in both classes (analysis; `scripts/sign/latent_geometry.py`, `lattice_p.py`): in the channels attention uses the relation a(+2) = 2 a(+1) holds on 0.98-1.00 of usage in 8/8 |.| runs (signed 0.99-1.00 in 7/8; angle tolerance 0.05 rad, a post hoc choice: at 0.3 rad the lattice shares change a lot), and the latent relation in 5/8 |.| runs, one of them through the mirror copy z_2 = -2 z_1 of E8.4 (cosine -0.98), against 7/8 signed. The unsolved |.| runs and 2 of the 3 unsolved signed runs have the relation in place (signed s1300: 0.68); what they miss lies outside the step code. There is no evidence of a sign cost without an inverse pair. Signed models on +1/+2 end with 0.92 of channels having both move angles of one sign (0.01 on +1/-1), the sign-coherent code Lemma 1 allows.

H5. Three times the ring budget (`RING_BUDGET_RESULTS.md`, seeds 40-47)

The |.| step at 900 epochs solves 3/8, as at 300 epochs on the same seeds (3/8); the solved sets differ.

H6. Cancelling share and ring strata

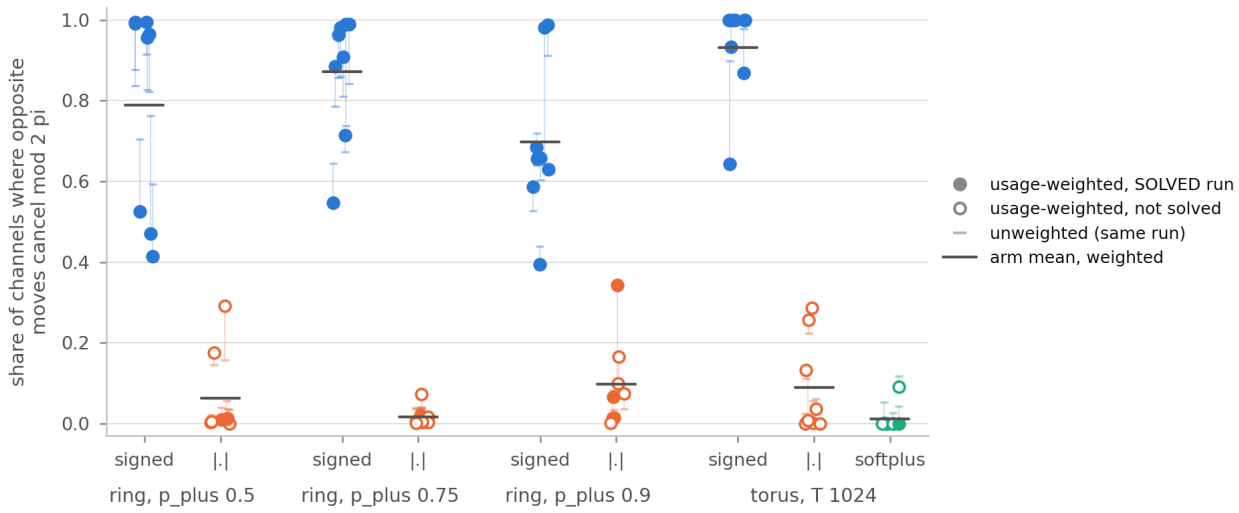


Figure H1 (`fig_wrap_share`; analysis). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens. Ring of 32 cells at three move biases (n = 8 each) and the torus at T = 1024 (n = 8). Bars: arm means of the weighted share. Signed steps 0.70-0.93; |.| steps 0.02-0.10; softplus 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)
ring p_plus 0.5	|.|	2/8	0.06 (0.00-0.29)	0.06 (0.02-0.16)

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)
ring p_plus 0.75	|.|	1/8	0.02 (0.00-0.07)	0.03 (0.02-0.04)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)
ring p_plus 0.9	|.|	4/8	0.10 (0.00-0.34)	0.06 (0.02-0.10)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)
torus T 1024	|.|	0/8	0.09 (0.00-0.29)	0.11 (0.00-0.27)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)

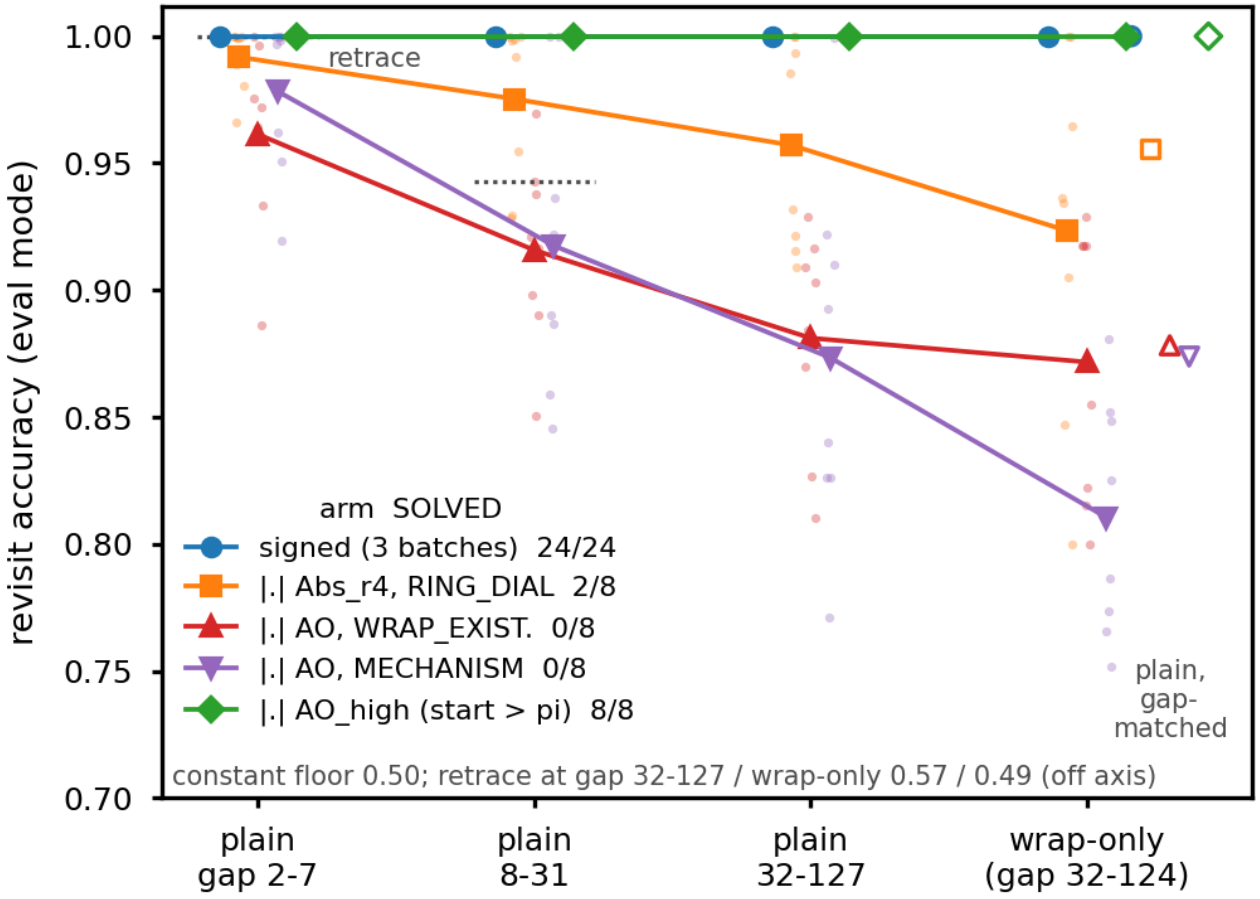


Figure H2 (`fig_ring_strata`; analysis). Revisit accuracy on the 1D ring by stratum (eval mode; 400 trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position; "wrap-only" revisits were seen only a lap away. Lines: arm means; dots: runs; hollow markers: plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 at gaps 2-7, 0.87-0.96 at gaps 32-127). Dotted: retrace within each stratum, 1.000 at gaps 2-7 and 0.943 at gaps 8-31 (0.57 and 0.49 in the long-gap and wrap-only strata, off axis); constant floor 0.50.

H7. Routes, measured (analysis; `docs/paper/expansion/scripts/sign/`)

End-state clocks (`clock_table.py` -> `clock_table_out.txt`: the clock / map ratio of E7.1', one value per run; the usage share on cancelling channels from `sign_clock.py`; `sign_clock.py`'s

one-sided map wrap(a_y), the old column, gave 1.683 / 1.632 / 1.549 / 1.394 / 1.520 / 1.219 for the |.| and frozen rows, and is reproduced by `clock_table.py` as a check):

batch, arm	SOLVED	clock / map, arm mean	usage on cancelling channels
ring MECHANISM, signed (default start)	8/8	0.018	0.96
ring MECHANISM, signed from the |.| start	7/8	0.015	0.96
ring GAUGE, signed (+1/-1)	8/8	0.016	0.95
ring MECHANISM, |.| (default start)	0/8	1.471	0.08
ring MECHANISM, |.| (|.| start)	0/8	1.571	0.02
ring GAUGE, |.| (+1/-1)	1/8	1.559	0.06
ring ACCOUNTS T2, signed, move latents frozen, |.| start	1/8	1.181 (the solved run 0.058)	0.15
ring ACCOUNTS T2, signed, move latents frozen, default start	7/8	0.167	0.87
ring MECHANISM, |.| from a large sub-pi start	2/8	1.423 (solved runs 0.10, 0.59)	0.20
ring MECHANISM, |.| from a start past pi	8/8	0.317 (wrapped; runs 0.11-0.76)	0.82
ring MECHANISM, signed from a start past pi	8/8	0.166 (runs 0.004-0.48)	--
torus TORUS_START, signed	8/8	0.001	0.77
torus TORUS_START, |.|	1/8	1.248 (the solved run 0.024)	0.16
torus TORUS_START, signed from a start past pi	2/8	0.737 (solved runs 1.39, 1.78)	--

"Clock-free at the end" holds for runs that succeed from a small start (arm means 0.001-0.018 here, 0.043-0.051 for the rank arms of G3); runs that succeed from a large start can end with a large ratio. In the two solved signed torus runs from a start past pi, 0.43 and 0.99 of the usage sits on channels whose per-pair clock is near pi (|wrap(k)| > 2.6 rad; `clock_table_out.txt`), where E7.1's bound says nothing; the independent audit found the second of them (s1206) at revisit accuracy 1.000 in all four parity strata of (n_-x, n_-y) on 16 fresh walks although the correct key's score flips sign with the parity (mechanism not identified). For unwrapped |.| channels the ratio is at least 1 by construction (E8.1).

The |.| cone inequality |k| >= |m| holds with zero violations on every sub-pi channel (an identity for |.|, listed as a check of the code). Trained signed ring models end on the collective zero set: |z_x + z_y| / (|z_x| + |z_y|) 0.044-0.053, cosine -0.994 to -0.995 (`latent_geometry.py`), MapFormer's antipodal end state (their Figs. 3d and 9b).

The cone boundary. Sub-pi |.| ring channels within 10% of the cone boundary (|delta| >= 0.9 |k| / 2, one move's step near zero): 0.30 (MECHANISM default start), 0.36 (|.| start), 0.25 (GAUGE), against 0.056 at initialisation and 0.00-0.02 for the large starts. |.| models end with 18.8-22.2 one-sided and 13.2-16.0 blind channels of 64 (signed 2.1-4.1 and 8.1-8.9; initialisation 15.6 and 5.0); these counts are unweighted over all 64 channels,

unused ones included. The MECHANISM initialisation values (0.056; 15.6 / 5.0) were recomputed with the script's functions by the independent audit; the script itself prints GAUGE's (0.078; 19.8 / 3.1). At fixed clock the largest map a |.| channel can carry sits on that boundary, so collapse onto the wall is the boundary optimum of a clock-map trade-off as well as the sticky kink of E4.2.

Latent-reachable share at initialisation (`latent_reach.py`, MECHANISM seeds 800-807). Share of the needed change of the angle sums that a latent move can produce: signed (to zero) 1.000 on every seed (E8.2(a)); |.| (to the nearest wrap) 0.339 unweighted and 0.561 usage-weighted (random-subspace baseline 0.125). The rest must come from per-channel readout changes. In the held constructed |.| models the wrap targets are carried by the readout (`aobal_geometry.py`): they cancel in 9-13 channels carrying 92-98% of usage, and a readout drawn at initialisation contains 0.3-11% of that target (random-subspace baseline 6%).

Straight paths with the content of a solved model fixed (`path_profiles.py`; ring, MECHANISM signed seeds 800-807; balanced gauge, so a straight line in W_out is a straight line in angle space; 64 fixed walks):

path	**loss at tau = 0, 0.2, ..., 1 (mean of 8)**	**accuracy at tau = 0 -> 1**	**largest rise above the running minimum, per seed**
signed, |default| start -> signed solution	9.46 9.18 8.65 7.50 6.41 0.00	0.30 -> 1.00	0.27 0.04 0.30 0.40 0.48 0.38 0.05 0.36
|.|, same start -> wrapped solution	9.46 8.23 7.87 7.96 7.44 0.00	0.30 -> 1.00	1.92 0.87 1.41 0.60 1.09 1.01 1.14 0.74
signed, collective latent route only	9.46 9.39 9.49 9.25 8.35 2.98	0.30 -> 0.72	0.03 0.00 0.00 1.36 2.15 1.44 0.21 0.02
|.|, the same latent route	9.46 9.41 9.66 9.90 9.68 9.21	0.30 -> 0.30	0.17 1.11 0.18 2.40 2.23 1.68 1.33 0.51

Every |.| seed's largest rise exceeds every signed seed's (complete separation, exact two-sided permutation p 1.6e-4).

The step map's basin with the content frozen (exploratory; `step_basin.py`, `basin_summary.py`; ring, seeds 800-807; content of the solved signed model frozen, only W_in, W_out and omega trained for 2000 steps of batch 32, about 2% of a full run; fresh-stream accuracy). These CPU runs are not deterministic and the step-map search is bistable: re-runs of single cells in the independent audit gave S_abs s801 0.944 -> 0.509 and 0.939, A_abs s803 0.548 -> 0.450 and A_high s804 0.921 -> 0.839, so the per-seed values and the "runs >= 0.9" column do not reproduce; only contrasts much larger than these swings are read:

start	**class**	**accuracy start -> end (mean)**	**runs >= 0.9**
default angles	signed	0.33 -> 0.83	3/8
|default| (the |.| start)	signed	0.29 -> 0.47	1/8
|default|, move latents frozen	signed	0.29 -> 0.40	0/8
|default|	|.|	0.29 -> 0.36	0/8
past pi, U(1.7, 4.5)	|.|	0.36 -> 0.64	1/8

start	class	accuracy start -> end (mean)	runs >= 0.9
halfway along the straight segment to the wrapped solution	|.|	0.32 -> 0.61	0/8

With the content fixed the start decides the step map's search (default vs |.| start, signed: +0.365, p 0.005; past pi vs small start, |.|: +0.280, p 0.0007; both survive Holm over the eight exploratory tests, two others at p 0.012-0.013 do not). The signed escape from the same-direction start that full training shows (7/8) is not reproduced by the step map alone (signed vs |.| from that start +0.104, p 0.32): in full training the content co-adapts. No run from halfway along the straight segment to the wrapped solution reached 0.9 (end accuracies 0.42-0.82), so within this budget the wrapped solution's basin did not extend that far; a re-run could change single runs.

H8. The sign route (`SIGNROUTE_RESULTS.md`, registered SUPPORTED; per seed in C10)

Six observation-step-free rank-4 arms on the ring at lr 1e-3 and 3e-3 (LRSWEEP's recipe), fresh seeds 1800-1807. The *hybrid* confines the readout to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, with the latent and hence the increment signed. *abs_tied* is |.| with the move latents tied antipodal, z_y := -z_x, so both moves step omega |w . z| and its code is a pure clock.

lr	signed	|.|	hybrid	abs_tied	primary (hybrid vs |.|, gated on signed vs |.|)
1e-3	8/8, 1.000	0/8, 0.918	8/8, 1.000	0/8, 0.739	Fisher p 0.0002; d +0.082
3e-3	8/8, 1.000	2/8, 0.958	8/8, 1.000	0/8, 0.748	Fisher p 0.007; d +0.042

Hybrid vs signed: d 0.000 at both rates. Descriptive end states (medians): hybrid u-ratio \| z_x + z_y\| / (\|z_x\| + \|z_y\|) 0.043 / 0.002 and cosine(z_x, z_y) -0.996 / -1.000, as signed (0.055 / 0.001; -0.994 / -1.000); |.| 0.661 / 0.664 and -0.133 / -0.165; the hybrid's effective readout is 0% negative and fully row-sign-coherent. Both the absence of the collective route (E8.3) and the far-side cancelling point (E8.1, E4.1) predict this outcome, since the hybrid restores both; the batch localises the cost to the increment and shows that non-negativity in the readout costs nothing.

H9. Exp and softplus increments (`SIGNROUTE_RESULTS.md`, registered breadth cells at fixed rates)

lr	arm	SOLVED	accuracy	d vs signed	label (Holm per test, 4 cells)
1e-3	exp	1/8	0.935	-0.065	signed learns better (Fisher 0.0056, perm 0.0006)
1e-3	softplus	1/8	0.879	-0.122	signed learns better (Fisher 0.0056, perm 0.0042)
3e-3	exp	1/8	0.958	-0.042	signed learns better (Fisher 0.0056; SOLVED test only; 5/7 unsolved runs still descending)
3e-3	softplus	3/8	0.966	-0.034	signed learns better (Fisher 0.0256; SOLVED test only; 2/5 unsolved still descending)

Both start every token at a positive step (mean move step at initialisation exp +1.051, softplus +0.706, signed +0.003), i.e. as index RoPE with a rescaled omega; no best-rate sweep was run for them.

I. CARoPE, CoPE and Selective RoPE

CARoPE (`model_mono_baselines.py`, `CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token; initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. Proposition 8 (E5): no channel can read every return. Registered (GENERALITY): torus 0.667 (0/8), below index RoPE (0.722, -0.055, p 0.040); ring 0.741 against index 0.730 (+0.011, not resolved).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1 in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p], table e at zeros initially; added to the attention logits; no rotary position. *Representability (constructed):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). The exact solutions form a lattice (P7 of v3: period 2 or 3).

batch	task	CoPE	SOLVED	comparison
GENERALITY (700-707)	ring	0.887	0/8	signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002)
WRAP_COPE (900-907)	ring, table at zeros / sinusoid past pi / below pi / slope-matched	0.840 / 0.913 / 0.867 / 0.943	1/8, 2/8, 1/8, 0/8	contrasts not resolved (bimodal seeds)
ACCOUNTS T4 (1100-1115)	256-cell line (never wraps) / ring	0.914 / 0.873	1/16 / 0/16	line: signed 1.000, 8/8 (Fisher p < 1e-4); index 0.772, 0/8

On both line and ring several CoPE seeds reach 0.99-1.00 held-out accuracy without meeting the loss criterion (at a held-out accuracy cut of 0.999 the line reads 5/16 and the ring 2/16; analysis). MapFormer v4 Table 2 reports CoPE failing 1D and 2D navigation (one layer 0.63 +/- 0.21 in 1D and 0.65 in 2D; four layers 0.88 and 0.76).

Selective RoPE (`SROPE_TORUS_RESULTS.md`, registered). Faithful Selective RoPE (Movahedi et al. 2026), the module reproduced from the authors' code to 6e-6, in the authors' best language-model configuration (SRoPE), the same without the SiLU (SRoPE_NoSiLU), the authors' state-tracking configuration (SRoPE_ST: low rank 32, no SiLU, no gate, a learnable bias of +1 rad per token per channel at initialisation) and our earlier generator swap (SRoPEGen), against MapWM (signed, rank 4) and index RoPE in one batch; torus, T = 1024, 900 epochs, seeds 1700-1707; own-stream floors blank 0.509, n-gram 0.578, retrace 0.845.

arm	accuracy T = 1024	SOLVED	classes	short / long / wrap-only	T = 2048
MapWM	1.000	8/8	all SOLVED	1.000 / 1.000 / 0.999	0.992
SRoPEGen	0.985	5/8	5 SOLVED, 2 DESC, 1 STAL	0.997 / 0.975 / 0.865	0.951
SRoPE_NoSiLU	0.969	2/8	2 SOLVED, 6 DESC	0.987 / 0.952 / 0.805	0.861
SRoPE (best LM config)	0.961	0/8	all DESCENDING	0.992 / 0.937 / 0.659	0.807
SRoPE_ST	0.769	0/8	7 STAL, 1 DESC	0.788 / 0.705 / 0.628	0.724
index RoPE	0.725	0/8	all STALLED	0.775 / 0.470 / 0.465	0.561

Registered: SRoPE vs MapWM fails to learn the map (d -0.039 [-0.050, -0.030], perm p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); SRoPE beats index RoPE (+0.236, p 0.0002); it uses its phase (angle set to zero: 0.961 -> 0.507, p 0.0078). Declared secondaries: the SiLU makes no detected difference (p 0.57); SRoPEGen vs MapWM not resolved (5/8 vs 8/8, p 0.20); SRoPE_ST fails against MapWM and is not distinguished from index RoPE (+0.044, p 0.23); at a loss threshold of 0.1 SRoPE vs MapWM is still 0/8 vs 8/8. Descriptive readouts: an observation's step is 0.86-0.96 of a move's in SRoPE / NoSiLU (0.45 SRoPEGen, 0.027 MapWM); opposite moves cancel (wrap-aware) in 0.75-0.80 of channel pairs (MapWM 1.00); the angles vary with context through the causal convolution (context CV 0.11-0.13, MapWM 0.00); SRoPE_ST's learned per-token angles have negative-mass share 0.00 for every token class and cancel in 0.00 of channels, so its bias keeps the step effectively non-negative.

J. Text world

Task (`TEXTWORLD_RESULTS.md`, `train_textworld.py`). The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (about 132 steps); held-out map; predict the observation word at revisits. A direction word appears only inside a movement clause (context-free steps). Floors: best constant 0.505, reversal-copy 0.597, retrace 0.934 (analysis; J6). Recipe: the torus recipe (A2), path MapWM r4 at one layer against index RoPE at one and two layers; 200 evaluation walks.

Step tables per seed (declared secondaries, `fig_textworld_out.txt`; "minus common" = opposition after removing the component shared by the four direction steps; ablation columns: held-out accuracy, eval mode without the dropout scale):

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
0	clock	1.730	0.0071	0.711	0.881	0.191	0.388
1	map	0.083	0.0029	0.041	0.998	0.893	0.505
2	clock	1.434	0.0064	1.144	0.999	0.665	0.482

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
3	clock	1.707	0.0022	0.480	0.995	0.673	0.463
4	clock	1.768	0.0052	0.501	0.999	0.539	0.445
5	map	0.120	0.0050	0.060	1.000	0.898	0.505
6	map	0.080	0.0117	0.040	1.000	0.950	0.505
7	map	0.119	0.0057	0.059	0.882	0.586	0.498

On the clock seeds the common component is parallel to the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 phase channels drift between visits to the same cell (2-4 on the map seeds). Synonym cosine 0.9996-1.0000 on every seed.

The paragraph and table above use the registered batch. The analyses below (`docs/paper/expansion/scripts/language/`) read the 8 registered text-world path runs (seeds 0-7) and the 8 MapWM, 8 NormStep and 8 DirOnly runs of TW_NORMSTEP (seeds 10-17) on a held-out map (np seed 424242, 40 walks of 1024 words unless stated). The manual forward used for the manipulations reproduces every model's logits exactly, and the role-annotated generator reproduces the task generator token for token. "Re-scored" = with the x1.111 attention-dropout scale (C9).

J1. The step table by word class (analysis; `tw_step_account.py`)

Steps omega-scaled, relative to the clause-level direction differential (one move):

word class	map-type models (14 learned-step)	clock-type models (10)
direction words (differential)	1.00 by normalisation; opposites cancel 0.001-0.013	same
adverbs, fillers	0.004-0.03	0.004-0.025
`and` / `a` / `she` (shared by fillers, seeing phrases, asides)	0.02-0.07	0.02-0.08
verbs, seeing words, object words, `.` (once per move)	0.02-0.14	0.24-1.49
aside-only words (`thought`, `about`, `remembered`, `story`)	0.07-0.16 per word	0.41-1.33 per word
aside sentence, net (usage-weighted; equal by construction to the aside noun's offset from its place)	0.05-0.12 of a move	0.04-0.16 of a move
DirOnly oracle: every non-direction word	0 by construction	--

The clause tick (what every move adds whatever its direction) is 0.03-0.23 of a move on map-type models and 3.3-6.8 moves (raw norm, all channels) on clock-type models, parallel to the verbs' step (cosine +0.999 to +1.000 on all 10).

J2. The tick predicts the drift; the clock counts moves (analysis; `tw_step_account.py`, `tw_clock_unit.py`)

Predicted drift per channel = mean |wrap(lag x tau_c)| over consecutive-visit pairs (median lag 8 moves, 1,221 pairs); observed = mean |wrap(dtheta_c)| at those pairs.

Classified at 1 rad they agree on 0.98-1.00 of channels on each of the 32 models. On the 22 map and DirOnly models both are near zero, so the agreement there is close to an identity; the informative evidence is the 10 clock models, where the predicted and observed drifting-channel counts match exactly. Clock models drift in 26-38 channels between consecutive visits (registered text world s0 / s2 / s3 / s4: 26 / 29 / 35 / 36; MapWM s11 / s13 / s14 / s15: 38 / 27 / 28 / 30; NormStep s11 / s16: 27 / 28); map models in 0-3. (The registered 31-38 of the CORRECTED block in `TEXTWORLD_RESULTS.md` counts drift from the first visit.) Regressing each move's net phase in the drifting channels on direction, aside present, filler count and adverb present: an aside sentence adds 0.009-0.062 of a tick, a filler word 0.001-0.007, an adverb word 0.001-0.007 (10 clock models). A word counter would give fillers about 1/8 of a tick each and a sentence counter would give asides about one tick: the clock counts moves. On all 10 clock models 2-6 channels tick by exactly pi per move (e.g. -3.141, -3.140 rad); on the even 64-torus a pi tick encodes the parity of the cell and is invisible to revisits (E7.2), and these channels carry up to 0.24 of retrieval usage. The registered finding that zeroing every non-direction step costs the clock seeds 0.33-0.69 and the map seeds 0.05-0.30 tracks the direction words' common component in the used channels (0.06-0.34 of a move on clock seeds, 0.01-0.03 on map seeds), which the other words cancel within the move (Spearman +0.905 across the 8 registered seeds, exact permutation p 0.0046; `tw_common_used.py`, p from `scripts/whatwhere/whatwhere_stats.py`; the used channels are those whose tick is 0 mod pi, a post hoc rule).

J3. The clock is idle (analysis; `usage_tick.py`, `tw_attention_asides.py`)

79-99% of retrieval usage sits in channels whose tick is 0 mod pi. Removing the drifting channels' q and k components from the score (re-scored accuracy):

model	intact	clock channels removed	as many map channels removed (5 draws)	only clock channels kept
text world s0 (eval mode 0.875, below ceiling)	0.973	0.717	0.43 [0.24, 0.58]	0.07
text world s2	0.999	0.990	0.25 [0.03, 0.63]	0.03
text world s3	0.998	0.993	0.06 (all 29 map channels)	0.06
text world s4	0.999	0.999	0.03 (all 28)	0.03
MapWM s11	0.989	0.985	0.48 (all 26)	0.48
MapWM s13	0.997	0.996	0.65 [0.50, 0.84]	0.48
MapWM s14 (eval mode 0.960)	0.995	0.902	0.25 [0.03, 0.72]	0.03
MapWM s15	0.995	0.994	0.07 [0.02, 0.23]	0.02
NormStep s11	0.998	0.996	0.46 [0.20, 0.60]	0.44
NormStep s16 (eval mode 0.835)	0.982	0.809	0.22 [0.04, 0.48]	0.04

On the 7 clock models at ceiling in eval mode the clock costs at most 0.009 to remove; the 3 below ceiling still lean on it (0.09-0.26).

J4. The used-channel ratio across data (analysis; `crossover_usage.py`, `usage_tick.py`, `tw_alpha_drift.py`)

Usage-weighted median of rho_c (E13) at the training length, all MapWM with a signed step:

data	used-channel rho	usage in clock-dominated channels	seeds
2D torus, 1 layer, matched length (SIGN_MATCHED signed)	0.064-0.079	0.000	8
text world, 1 layer (23 of 24 learned-step models; the unconverged tw s0: 2.5)	0.06-0.23	0.01-0.30	24
code, bytes, 9 layers (trained at 2048)	4.5-6.6	0.50-0.55	3
TinyStories, words, 6 layers (trained at 512)	9.9-12.3	0.42-0.43	3
k-back recency with fillers, 1 layer (signed rank 4)	21.8-22.1	0.93-1.00	8

Clock-dominated: rho_c > 1 and more than half a turn per training window. The pooled growth exponent alpha does not separate map from clock data: it reads 0.61-0.74 on text-world map seeds (the true walk: 0.668), 0.98-1.02 on text-world clock seeds, which carry a perfect map, 1.00-1.01 on TinyStories and 0.99-1.01 on code. The used-channel rho separates the clock seeds (0.07-0.23) from TinyStories (9.9-12.3).

J5. Asides: the offset is what beats the oracle (analysis; `tw_attention_asides.py`, `tw_attention_asides_dironly.py`)

`TW_NORMSTEP_RESULTS.md` (registered) found that DirOnly, whose steps are restricted to the twelve direction words, stops at 0.970-0.974 on 8/8 seeds because an aside's noun sits at the place's phase, and that learned-step models make none of these errors; re-scored, every learned-step arm beats DirOnly (-0.024, p 0.0002). Two manipulations at evaluation, each keeping the phase after the aside unchanged:

- *Collapse* (learned models): every aside word is put at its clause's object-slot phase. The 19 learned-step models at ceiling in eval mode fall from 0.989-1.000 to 0.952-0.975, and 71-100% of their errors then name an object told in an aside at that cell (before: 0-13 errors, one of that kind, in MapWM s13). The 5 models below ceiling respond partly (0.966-0.994). The 19 / 5 split at eval-mode accuracy 0.98 is a post hoc cut (it falls in a gap, 0.960 against 0.989).
- *Lift* (DirOnly): intact, DirOnly scores 0.962-0.981 on this stream and 100% of its errors are aside errors on 8/8 seeds. Offsetting its aside words by half a move in-plane gives 0.999-1.000 on 8/8; by one move off the map plane 1.000 on 7/8 (one seed 0.972); by a quarter move off-plane 0.963-0.996 (a variant added after the main run). The lift sizes are free choices, not fitted.

At each revisit query the 19 learned-step models at ceiling put 0.55-0.76 of their attention on the object slot of the most recent visit to the cell and 0.004-0.035 on aside nouns told there; DirOnly puts 0.041-0.077 on those aside nouns.

J6. Retrace on the text world (analysis; `tw_retrace_floor.py`, `tw_strata_200.py` -> `tw_strata_200walks_out.txt`)

The retrace rule of B, ported to the text world, scores 0.9337 on the registered evaluation stream (0.9365 on TW_NORMSTEP's), where the constant and reversal-copy rules reproduce the registered 0.5054 / 0.5965. On all 200 walks of that stream (6,381 revisits) the rule applies to 5,525 (86.6%) and is never wrong when it applies; on the other 856 (13.4%), places reached by another route than undoing the last run, it predicts 'nothing', which is right on the 50.6% of them whose cell is blank. Accuracy by stratum (mean over 8 seeds; fresh seeds 2-7 in brackets):

model (registered batch, n = 8)	all	retrace applies (5,525)	retrace cannot reach (856)	... cell holds an object (423)	... blank cell (433)
path, 1 layer, re-scored	0.993	0.999	0.956 (0.978; min 0.786 at s0)	0.925 (0.967; s0 0.617, s7 0.872)	0.985
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)	0.984
index RoPE, 2 layers	0.772	0.828	0.411 (0.34-0.55)	0.129 (0.09-0.15)	0.687
index RoPE, 1 layer	0.505	0.513	0.454	0.004	0.894
best constant	0.505 ('nothing')	--	0.506 ('nothing')	0.088 ('cat')	1.000 ('nothing')

The object-cell column is the stratum "retrace wrong" of the first version of this analysis (`tw_strata.py`, first 100 walks: 5.9% of 3,055 revisits, path 0.920 / 0.961 fresh, RoPE 2L 0.138, constant 0.112). Its best constant excludes 'nothing' by construction, and one index layer scores near zero there because it predicts 'nothing'. On the whole non-retrace stratum, which needs no outcome-defined split, two index layers score below the constant. Two-layer index RoPE learns part of retrace (its best seeds reach 0.927 and 0.926 overall, against retrace's 0.934) and nothing that needs the map. Seeds at ceiling score 0.936-1.000 on the object-cell stratum.

K. What and where; the leak

K1. What and where at T = 1024 (analysis; `docs/audits/2026-10-20/WHAT_WHERE_T1024.md`, `causal_whatwhere_T1024.py`)

The causal what/where test re-run in the regime where a map is needed: the SIGN_MATCHED checkpoints (torus, trained and tested at T = 1024, one layer, rank 4, n = 8 per arm; floors on these walks blank 0.506, n-gram 0.575, retrace 0.843). The rebuilt forward equals the model logits exactly on all 32 checkpoints; intact accuracy matches the registered evaluation to <= 0.0007. Held-out accuracy when the score is rebuilt:

condition	signed (learns the map)	|.|	softplus	index RoPE
intact	0.998	0.821	0.780	0.731

condition	signed (learns the map)	|.|	softplus	index RoPE
token identity within type removed from the score (token type kept)	0.966 (-0.032; 8/8 lower, sign-flip p 0.008, uncorrected)	0.544 (-0.277)	0.476 (-0.305)	0.479 (-0.252)
one shared position kernel x per-pair content gain plus offset	0.998 (-0.000)	0.584 (-0.237)	0.675 (-0.106)	0.439 (-0.293)
additive position + content	0.895 (-0.104)	0.350	0.512	0.492
token type removed too (one global mean)	0.113	0.295	0.302	0.399
content only (no rotation)	0.505	0.399	0.385	0.490
SS(P) / SS(S): within-row score sum of squares carried by the content-averaged score	0.925	0.644	0.577	0.618
content x position interaction share	0.061	0.187	0.174	0.353

The model that learns the map factorises what and where: one position kernel per head, scaled by a per-pair content gain plus offset (2 x 441 parameters per head, fitted on the calibration walks, so this condition nests the additive one), reproduces it exactly; removing token identity within type (which also removes action identity within the action class and blank against object within the observation class) costs 0.032 (8/8 seeds lower; sign-flip p 0.008, the design's minimum; paired t p 0.012; Holm over the 24 contrasts 0.08; the effect sits at the MDE of 0.031); token type is essential. The failing models do not factorise: removing identity costs 0.25-0.31 on average (|.|: 7 of 8 runs lose at least 0.15; Abs_r4 s0 loses 0.020), and a shared kernel times a gain does not restore them. Within the non-negative arms, across 16 runs, accuracy correlates with the share of it that survives removing token identity (Spearman 0.84, p 5e-5; a post hoc statistic, `scripts/whatwhere/whatwhere_stats.py`); the two non-negative runs that reach > 0.95 keep 0.97 and 0.99 of their accuracy, the four lowest softplus runs fall to 0.03-0.07 (5-12% of their accuracy), and the |.| runs keep 0.22-0.77 of theirs except Abs_r4 s0 (0.98 of 0.931). (Intact softplus reads 0.780 on this rebuilt forward against the registered 0.781. The `.txt` of the source was written by `--report` after the run; the JSON is the run's.)

K2. What and where at T = 128 (analysis; `docs/WHAT_WHERE_CHECKS.md`; the 48 PAPER2X2 checkpoints)

score used	RoPE	PoPE	MapWM r2	MapWM r4	MapPoPE r2	MapPoPE r4
intact	0.805	0.679	0.971	1.000	0.999	1.000
position only (object identity removed, token type kept)	0.463	0.476	0.837	0.989	0.974	0.986
shared position kernel x content gain	0.430	0.435	0.791	1.000	0.988	1.000

On short walks, where retrace already scores 0.934, converged path models keep 0.974-0.989 with object identity removed and 0.988-1.000 with a shared kernel times a content gain; index models fall below the n-gram floor (0.598). Separation as such is MapFormer's (their Fig. 9).

K3. The what-to-where leak (`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`, registered)

On a task with a fresh map and 16 fresh objects per sequence (32 x 32 torus, T = 1024, rank 4), object tokens leak a small step into the phase. Removing the step from object tokens (ActOnly) or normalising the step input (NormStep: the step reads LayerNorm(embedding)) lets the model converge: 16/16 SOLVED (unseen-object accuracy 0.9997 / 0.9998) against 0/8 for MapWM (0.9890, still descending). In distribution the remedies gain +0.0107 in unseen-object accuracy (p 0.0002, 8/8), equal to MapWM's leak; the accuracy and loss are strongly coupled there (r(loss, accuracy) -0.94). On fresh seeds the convergence result replicated (NormStep 8/8 against MapWM 0/8) and the accuracy gain did not (+0.0045, p 0.46; 0.9940 against 0.9895). A score change alone leaves the leak in place (GAIN_PHASE).

L. PoPE's score

L1. Registered

batch	result	file
PAPER2X2 (T = 128, r2, n = 8)	RoPE 0.805, PoPE 0.679, MapWM 0.971, MapPoPE 0.999; interaction +0.154 (8/8); at r4 +0.125 with both path arms at 1.000	`PAPER2X2_RESULTS.md`
MAPPOPE_PAIR (T = 128, r2, n = 16)	the score rule, not the doubled angle count: +0.0243 (p 0.012; SOLVED 16/16 vs 10/16), plausibly convergence speed, +0.009 on fresh seeds; the angle count +0.0002 at a ceiling (0.9995-0.9997), uninformative	`MAPPOPE_PAIR_RESULTS.md`, `GAIN_GRAIN_RESULTS.md`
SCORE_RANK (T = 1024)	no rank-2 rescue: 3/12 vs 3/12 (App. G2)	`SCORE_RANK_RESULTS.md`
DELTA_FREE (T = 128, r2, n = 8, fresh seeds 1600-1607)	PEAK-PINNING SUPPORTED: freeing delta raises the index model by +0.122 [90% CI +0.076, +0.167] (8/8, sign-flip p 0.0078), concentrated in multi-step reversals (R3+ +0.180, Holm p 0.031), and leaves the path model at ceiling (-0.018, p 0.31; SOLVED 6/8 vs 8/8, Fisher p 0.47)	`DELTA_FREE_RESULTS.md`

DELTA_FREE arms (accuracy at T = 128 / 512 / 1024, SOLVED): PoPE-Flat (index, delta pinned: zero initialisation held at 0 by the clamp's dead zone) 0.649 / 0.615 / 0.599, 0/8; PoPE-FreeDelta (delta initialised U(-2 pi, 0), clamp removed) 0.771 / 0.536 / 0.442, 0/8; MapPoPE-Flat (path, pinned) 0.999 / 0.980 / 0.933, 8/8; MapPoPE-FreeDelta 0.981 / 0.901 / 0.831, 6/8; MapPoPE-FrozenDelta (path, delta frozen at the free arms' draw) 0.999 / 0.929 / 0.832, 8/8. The delta x phase interaction is -0.140 [-0.186, -0.092] (0/8 seeds positive). Declared secondaries: with delta pinned 0% of index content pairs prefer offset 3 over offset 1 (kernel argmax at offset 1 in 1074/1088 pairs), freed 50-100% per seed with argmax spread over offsets 3, 4, 7, 8; attention on the newest observation (never the target) 0.287 -> 0.003; index gain by stratum R1 +0.039 (n.s.), R2 +0.117, R3+ +0.180, unvisited-run +0.014; path target alignment C 0.987 pinned vs 0.616 free and 0.614 frozen, with the code term C0 0.993 / 0.798 / 0.884; pinned arms 96-100% in the dead zone; free arms moved mostly by weight decay. Beyond the training length the path robustness contrasts (free or frozen minus pinned at T = 1024: -0.101, -0.102) are

below the registered threshold, and the free index arm's retrace kernel does not extrapolate (-0.158 at T = 1024).

L2. The mechanism, probed (analysis; `docs/paper/expansion/scripts/mappope/kernel_probe.py, probe_stats.py`)

Layer 0 rebuilt by hand; every run's logits equal the model's own (difference 0.0); per-seed accuracies reproduce the stored evaluation. T = 128, PAPER2X2 checkpoints, 100 walks per seed:

readout	RoPE	PoPE	MapWM r2	MapPoPE r2
target coherence C (path arms)	--	--	0.703	0.995 (p 0.0002, 8/8)
code term C0 (path arms)	--	--	0.905	0.996 (p 0.0002)
same-cell attention mass (path arms; retrieval head)	--	--	0.747	0.925 (p 0.0008)
attention on action keys (path arms; retrieval head; over both heads 0.223 vs 0.147, p 0.15)	--	--	0.179	0.041 (p 0.0002)
content pairs with K(3) > K(1) (index arms)	53-100% per seed	0% on every seed	--	--
attention on the newest observation (index arms; the more recency-weighted head)	0.002	0.307 (p 0.0002)	--	--
R3+ accuracy (index arms)	0.779	0.566 (p 0.0002)	--	--

The argmax of the content-pair kernel over the odd offsets 1-15, where an observation key can sit, is at offset 1 for 1068/1088 PoPE pairs; for RoPE it is at offsets 3 (569), 7 (342), 11 (156) and 15 (21): every RoPE pair peaks on a retrace offset 4j - 1, a comb (`kernel_argmax_odd.py`; the argmax over all offsets 1-16 also lands on even offsets, where no observation key sits). The PoPE - RoPE accuracy gap decomposes over retrace strata as R1 -0.006, R2 -0.005, R3+ -0.110 and non-retrace -0.004. MapWM's content phase costs C0 - C = +0.20 (7/8 seeds; a statement about target coherence on the retrieval head, not an accuracy prediction). About 80% of the +0.154 interaction is the index side, (MapPoPE - MapWM) - (PoPE - RoPE) = (+0.028) - (-0.126).

T = 1024, SCORE_RANK checkpoints (r2 n = 12, r4 n = 8; 40 of the 100 registered walks per seed; each stratum read on the head with the most same-cell attention in that stratum, `HEAD_RULE=stratum` -> `kernel_probe_score_rank_hstratum_out.txt`, `probe_stats_hstratum_out.txt`; the first version read every stratum on the head chosen over all revisits):

stratum	MapWM r2: C / C0	PoPE score r2: C / C0	MapWM r4: C / C0	PoPE score r4: C / C0
plain gap < 128	0.617 / 0.900	0.960 / 0.960	0.926 / 1.000	0.997 / 1.000
plain gap >= 128	0.538 / 0.795	0.908 / 0.905	0.923 / 0.993	0.991 / 0.993
wrap-only	0.144 / 0.399	0.485 / 0.482	0.904 / 0.967	0.969 / 0.970

On short gaps the score raises C by +0.34 (p 0.0002) and C0 by +0.06 (0.900 vs 0.960, p 0.16): the repair is mostly in the readout term. On gap >= 128 revisits the score raises C (0.538 to 0.908, p 0.005) and leaves C0 low (0.80 / 0.91, p 0.32, against 0.99 at rank 4); on wrap-only revisits C0 is 0.40 / 0.48 under the two scores (p 0.64), and rank 4 raises it to 0.97 under either (MapWM +0.568, p 0.0009; PoPE score +0.488, p 0.012). In both rank-4 models the wrap-only revisits are carried by the head that is not the short-gap head (C0 0.97 there). At T = 128 MapPoPE's C0 is also higher than MapWM's (0.996 vs 0.905), so on short walks a better readout and a better code co-occur. About 60 contrasts are computed in this appendix without a multiplicity correction; the p 0.0002 contrasts survive Holm over the family, those at p 0.03-0.04 (e.g. MapPoPE - MapWM short-gap accuracy +0.114, p 0.036, consistent with the registered +0.115 on the same checkpoints) do not.

M. Loops and depth on a rank-2 code

M1. Registered (`LOOP_RANK_RESULTS.md`; torus, T = 1024, 900 epochs, n = 8, one batch)

arm	parameters	passes	SOLVED	accuracy	vs A: Fisher / permutation
A: r2, 1 layer	204,373	1	0/8	0.894	--
L: r2, looped x4	204,373	4	2/8	0.973	0.47 / 0.0034 (+0.079, CI [+0.026, +0.131])
L4: r2, 4 real layers	799,189	4	5/8	0.990	0.0256 / 0.0012 (+0.096)
C: r4, 1 layer	204,757	1	8/8	0.998	reference

L and A are identical in parameters and initialisation at every seed (maximum weight difference 0.0). Holm-adjusted p 0.0034 / 0.0028 / 0.0028; L4 - L +0.017 (p 0.0009). The registered verdict is UNMEASURED (it required L to solve >= 4/8). Final losses form three regimes: A 0.051-0.675, L 0.045-0.127, L4 0.011-0.070, C 0.002-0.010. Strata (short / long / wrap-only, registered secondaries): A 0.950 / 0.671 / 0.557; L 1.000 / 0.927 / 0.746; L4 1.000 / 0.990 / 0.887; C 1.000 / 0.995 / 0.977.

M2. Per pass (analysis; `scripts/recurrence/pass_strata.py`, `attn_offsets.py`)

The re-implemented forward matches each model's own forward and its stored evaluation (max |diff| <= 3.1e-4). L at `n_loops = j` is exact; L4's intermediate readouts are a logit lens.

arm	stratum	pass 1	2	3	4	5	6	8
L	all	0.381	0.887	0.964	0.973	0.971	0.954	0.847
L	long	0.363	0.612	0.875	0.927	0.921	0.871	0.731
L	wrap-only	0.393	0.519	0.700	0.747	0.730	0.699	0.623
L4	all	0.507	0.575	0.935	0.990			
L4	wrap-only	0.504	0.507	0.601	0.887			

One pass of L's trained weights scores below the always-blank floor (0.381 vs 0.51; mean, per seed 0.09-0.67, 7 of 8 below); the loop is a fixed-depth computation, and past four passes accuracy falls. Same-cell attention mass (best head per seed, `pass_strata_besthead.py` -> `out/PASS_STRATA_BESTHEAD.txt`; head labels are a symmetry, and at pass 2 the retrieval head is head 1 on three seeds) rises over L's passes 0.03 -> 0.50 -> 0.64 -> 0.63, and the share of queries whose top-attended key is a same-cell observation 0.35 -> 0.86 -> 0.88 -> 0.92: retrieval forms at passes 2-3 and pass 4 sharpens it. Attention entropy falls 6.1 -> 2.4 nats (head mean, 10 walks per seed; `attn_offsets.py`). On wrap-only revisits L4's last layer puts its top attention on a same-cell observation 81% of the time (best head), from a code whose own one-pass reader finds the cell 12% of the time (the reader's top-1 cell, a different quantity). Mass on same-cell action tokens stays at 0.01-0.15 (best-head means over passes 1-4). With an index code every pass works on short gaps only: the index loop's long-gap and wrap-only accuracy sits at 0.50-0.52 after every pass while short gaps climb 0.41 -> 0.94 (`index_loop_passes.py`, FACTORIAL checkpoints).

M3. Code quality, read without the block (analysis; `phase_reader_soft.py`, `phase_code.py`)

The one-pass reader of E11.1 (token-type phase offsets on query and key, a key bias, a soft vote over observations), fitted to each model's own theta on 24 separate walks and scored on the 50 evaluation walks:

arm	model (all / short / long / wrap)	one-pass reader	model - reader
A	0.894 / 0.950 / 0.671 / 0.558	0.866 / 0.940 / 0.558 / 0.448	+0.028 / +0.010 / +0.113 / +0.110
L	0.973 / 1.000 / 0.927 / 0.747	0.893 / 0.971 / 0.608 / 0.406	+0.080 / +0.029 / +0.319 / +0.341
L4	0.990 / 1.000 / 0.990 / 0.887	0.897 / 0.967 / 0.653 / 0.453	+0.094 / +0.033 / +0.337 / +0.435
C	0.998 / 1.000 / 0.995 / 0.977	1.000 / 1.000 / 1.000 / 0.997	-0.002 / 0.000 / -0.005 / -0.020

Code quality L - A +0.027 (permutation p 0.28), L4 - A +0.031 (p 0.27), both below the exact-t MDE (0.074 / 0.080; long-gap +0.050 / +0.095 against 0.27-0.28; `reader_mde.py` -> `out/READER_MDE.txt`): whether the loop and depth learn a code one pass reads better is unmeasured; C - L +0.107, C - L4 +0.103 (p 0.0002 each). The model is scored on 100 walks and the reader on the first 50 of them; on matched walks the margins change by <= 0.004. The reader under-fits on some runs: with five times the fitting steps on seeds 0, 1 and 3 the independent audit found L's long-gap reader accuracy up by 0.03-0.06, so L's long-gap margin (+0.32) is probably 0.03-0.05 too large; no verdict changes. Model-minus-reader margin against A's: L overall p 0.0036, long 0.0025, wrap-only 0.0002; L4 0.0031, 0.0045, 0.0002. A simpler top-1 kernel reader gives the same verdicts.

M4. Step-map geometry (analysis; `geometry.py`)

Per run, the omega-weighted x- and y-step phase vectors; *2D map*: common mode below 0.1 of the map step and the second singular value above 0.3 of the first; *1D map*: common

mode below 0.1, x and y steps collinear (sigma_2 / sigma_1 <= 0.002 in every such run); *drifting*: common mode >= 0.1.

arm	2D map	1D map	drifting
A	1/8	0/8	7/8
L	0/8	4/8 (code x + kappa y, |kappa| 1.38-1.42)	4/8
L4	1/8	4/8 (|kappa| 0.62-1.72)	3/8
C	8/8	0/8	0/8

2D map, aids vs rank 4: 1/16 vs 8/8 (Fisher p 1.2e-5); the aids do shift the class, 1D cancelling 8/16 against 0/8 for plain rank 2 (Fisher p 0.022, post hoc), so they change which code is learned within the family but rarely reach rank 4's. The class thresholds are post hoc and stable (any common-mode cut in (0.08, 0.34) and any second-singular-value cut in (0.002, 0.57) give the same classes; closest call C s0 at 0.08). In the reader's terms the 1D code is the collapsed end state of a rank-D head with the common mode removed. At initialisation the loop multiplies the step map's gradient norm by 2.9 and leaves its batch-to-batch coherence unchanged (0.46 -> 0.41; `grad_at_init.py`): the loop does not deliver a stronger or more consistent signal to the code, and Adam removes the scale.

N. EM's recency deficit: a contrast case

Single-origin MapEM (one p_0, an ablation of the paper-faithful model) on k-back recency (T = 1024, 300 epochs):

leg	result	file
exists	the rewind installed and frozen scores 1.000 on 8/8 at T = 1024 and 2048	`WARM_RESULTS.md` (W1)
kept	installed at the 1/64 scale and trained: 0.642; at 8x the scale 0.941 (4/8 >= 0.95); fully kept (1.000, 8/8) with `w_in`'s content columns held at zero	`WARM_RESULTS.md`, `NOLEAK_RESULTS.md`
from scratch	0.600, against 0.975 for MapWM and 0.837 for paper-faithful MapEM	`RECENCY_EM_RESULTS.md`
found per token	with one offset k every query shares, EM finds a 63-symbol rewind on 7/8 seeds (0.985), faster than MapWM	`SEARCH_RESULTS.md`
budget	four times the budget moves the 64-offset task from 0.578 to 0.928 (6/8 near 1.000)	`SPREAD_RESULTS.md` (an arm of a registered batch; the budget contrast itself was not registered)

EM's deficit is a search problem of a different kind: exposure per query token, which budget closes. Rank 2, by contrast, solves 0/8 with twice the budget on the torus (5/8 runs still descending at 1800 epochs), and |.| on the ring 3/8 at three times the budget (`RING_BUDGET_RESULTS.md`).

O. Analysis scripts

Every number marked (analysis) comes from one of these scripts (CPU, eval-only, on stored checkpoints of registered batches unless stated). "Verification" = the check the script runs before any new readout. Independent audit: every script below was audited results-blind and re-run on CPU (`docs/paper/review_draft5/ANALYSIS_SCRIPT_AUDIT.md`); every committed output re-derived, except `sign/step_basin.py`, which is not deterministic (its numbers are labelled exploratory). The audit's corrections are applied in this version; the scripts marked "new" were added to apply them, so that every quoted number has a committed generator.

script (under `docs/paper/expansion/scripts/`)	what it computes	used in	verification
`depth/torus_depth.py`, `summarize_torus.py` (`--segments`: new)	torus revisit classes, runs spanned, attention masses; fixed-gap segments and runs spanned by long-gap revisits	Table 3; Sec. 6; F1-F2	stored accuracy to 4 decimals (16 runs); forward = model logits
`depth/torus_probe.py`	ridge probes of the residual stream	F3	same stream
`depth/torus_runlen.py`	run-length-shifted evaluation	F4	-- (distribution-shifted)
`depth/ring_depth.py`, `summarize_ring.py`	ring classes, probes, one-layer mixture test	F5	stored accuracy for all 128 runs (< 2e-3)
`depth/ring_probe_window.py`, `ring_class_floors.py`, `ring_plain_bins.py` (new)	late-position probes and old-displacement targets; per-class floors; plain-only run bins	Sec. 6; F5	same stream as `ring_depth.py`
`depth/dyck_distance.py`, `dyck_crosstab.py`	Dyck closing accuracy by blocks and distance	F6	arm means equal the registered ones
`sign/clock_table.py` (new)	the clock / map ratio of E7.1' for every sign and rank arm, per run and per head	Table 4; Sec. 7.1; G3; H7	reproduces `sign_clock.py`'s one-sided ratios as a check
`sign/sign_clock.py`, `rank_clock.py`	cancelling usage share; observation and latent clocks	G3; H7	cone inequality as an identity (0 violations)
`sign/latent_geometry.py`, `lattice_p.py`	latent end states; +1/+2 relations	H4; H7	initialisation rebuilt bit-exact
`sign/latent_reach.py`, `aobal_geometry.py`	latent-reachable share; readout targets	H7	--
`sign/prop_checks.py`	numerical checks of E8.3-E8.4	E8	--
`sign/path_profiles.py`	loss along straight paths	H7	--
`sign/step_basin.py`, `basin_summary.py`	step-map-only training from fixed content (CPU; not deterministic: exploratory)	Sec. 7.3; H7	exact solutions held (pilot)

script (under docs/paper/expansion/scripts/)	what it computes	used in	verification
rank/rank_geom.py, sweep_final.py, summarise_final.py, c1_runlevel.py	off-plane share, readgain, r_beta	G4	drift and frame conditioning equal the registered head statistics to 4 decimals
rank/escape_start.py, init_frame.py, grad_hiding.py	escape re-read; initial conditions	G4; G6	--
rank/clock_decomp.py, c1_runlevel_once.py (new)	move / observation parts of the leftover clock; escape-arm clock sizes; C1 counting each network once	Sec. 7.2; G4	--
rank/train_snap.py, analyze_snap.py, train_nd_cpu.py	CPU pilots on seeds 3000-3002, 4000-4002	G6	outside every registered batch
recurrence/pass_strata.py, attn_offsets.py, index_loop_passes.py; pass_strata_besthead.py (new)	per-pass accuracy and attention; best head per seed	M2	forward and stored evaluation to <= 3.1e-4 (1e-6 on FACTORIAL)
recurrence/reader_mde.py (new)	MDE of the reader's code-quality contrasts	Sec. 7.4; M3	--
recurrence/phase_reader_soft.py, phase_code.py	one-pass reader fits	Sec. 7.4; M3	fitted on separate walks
recurrence/geometry.py, training_curves.py, grad_at_init.py	step-map classes; curves; gradients at initialisation	M4	--
mappope/kernel_probe.py, probe_stats.py (HEAD_RULE=stratum / SUFFIX=_hstratum: new); kernel_argmax_odd.py (new)	kernel, coherence C and C0, attention by offset; per-stratum head; kernel argmax over odd offsets	Sec. 7.4; L2	rebuilt logits equal the model's (0.0); per-seed accuracies reproduce the stored evaluation
language/tw_lib.py and the tw_*.py scripts; tw_strata_200.py (new: all 200 walks, the non-retrace stratum and its floors)	step table, drift, clock unit, asides, attention, retrace, strata	Table 7; Sec. 8; J1-J6	manual forward logit-identical; generator token-identical; N_WALKS=100 reproduces tw_strata.py
language/usage_tick.py, crossover_usage.py, tw_alpha_drift.py	used-channel rho; growth exponent	Sec. 9; J4	--
docs/audits/2026-10-20/causal_whatwhere_T1024.py; whatwhere/whatwhere_stats.py (new)	what/where at T = 1024; its sign-flip p, SS(P)/SS(S) and Spearman, and the text-world Spearman p	Sec. 8; Sec. 9; J2; K1	rebuilt forward exact on 32 checkpoints; intact accuracy within 0.0007

Post hoc choices (rule 7; from the audit's Sec. 5): the best-head rule of the kernel probes and the per-stratum head rule that replaces it at T = 1024; 40 of the 100 registered walks at T = 1024 (L2); the r_beta < 0.5 cut, the 10x criteria for t_kappa / t_lam and the off-plane share as the escape statistic (G4, G6); geometry.py's class thresholds (M4, stable over wide ranges); sign_clock.py's 0.1 / 0.05 rad cuts and the near-pi cut 2.6 rad of clock_table.py (H7); the lattice_p.py tolerance (H4); the path-profile rise statistic (H7); the depth run-length bins (F1, F4); the text-world 0.98 ceiling cut, the blank exclusion of the object-cell stratum and the lift sizes (J5, J6); the what/where Spearman (K1); the 0 mod pi

used-channel rule (J2-J4). The audit found that none of its corrections depends on a threshold choice.